



Dall'AI Discriminativa all'AI Generativa

by Daniele Grotti

TABLE OF CONTENTS

01

Definizioni

02

Machine Learning

03

AI Discriminativa

04

AI Generativa

05

Evoluzione degli
LLM

06

Q/A

Chi sono

- **Innovation Manager:** aiuto le aziende a navigare nel mondo digitale e a implementare nuove tecnologie legate all'intelligenza artificiale e all'automazione. Il mio lavoro consiste nel facilitare la collaborazione e l'integrazione delle idee innovative in tutti gli aspetti delle organizzazioni, dalla ricerca e sviluppo al marketing, dalla produzione alle vendite.
- **Data Scientist:** offro consulenza esterna per la creazione di pipeline per la pulizia dei dati, la trasformazione e la modellazione. La mia specialità è la consulenza strategica: utilizzo dati finanziari, di produzione ed open data per ottimizzare le strategie aziendali. Offro consulenza sulle opportunità di business, identificando aree di miglioramento e suggerendo soluzioni innovative.
- **Machine learning professor:** mi occupo di formazione post-laurea per il progetto Big Data Lab della regione ER dal 2018. Sono docente di corsi di programmazione Python e di algoritmi di ML e Deep Learning. Creo percorsi di formazione personalizzata per dirigenti e manager, aiutandoli a migliorare le loro competenze in questo campo in rapida evoluzione.



SCAN ME

 futura AI

Definizione di dati

I dati sono rappresentazioni originarie, cioè non interpretate, di un fenomeno, evento, o fatto, effettuate attraverso simboli o combinazioni di simboli, o di qualsiasi altra forma espressiva legate a un qualsiasi supporto.

- Dati sono rappresentazioni di eventi o fatti
- Non interpretate (originarie)
- Attraverso simboli (o combinazioni di simboli)
- Contenute su supporti (forma espressiva)

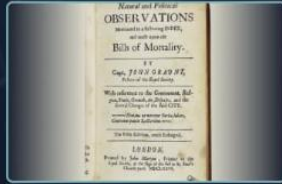
HISTORY OF DATA

19,000 BC



The Ishango bone holds the first evidence of data collection and storage.

1600s



John Graunt introduces the concept of data analysis in 1663.

1800s



Herman Hollerith designs a machine that helped complete the US census in 1890.

1900s

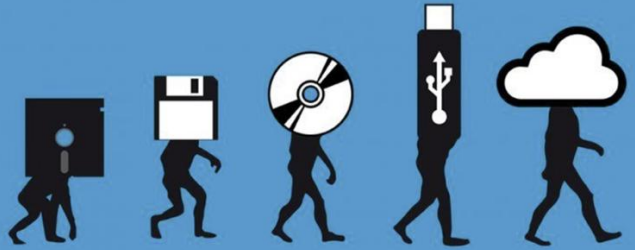


Fritz Pfleumer invents the magnetic tape which later inspired the invention of floppy disks and hard disk drives.

1990s

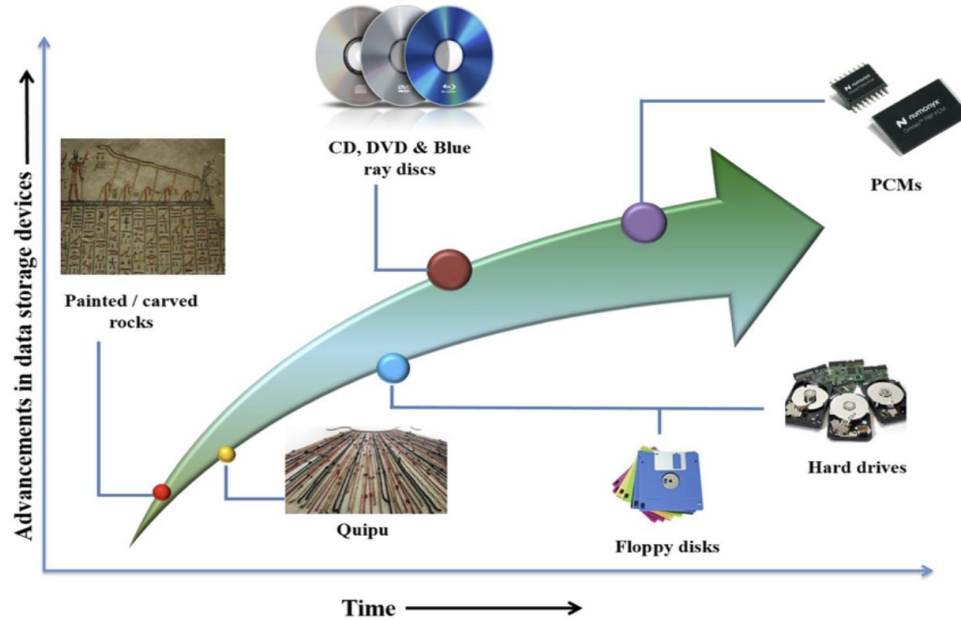


Sir Tim Berners-Lee invents the World Wide Web.



A sequence of five silhouettes of a person walking, each carrying a different storage device on their back: a large square tablet, a floppy disk, a CD, a USB drive, and a cloud icon.

THE EVOLUTION OF STORAGE DEVICES



Definizione di Informazione

L'informazione deriva da un dato, o più verosimilmente da un insieme di dati, che sono stati sottoposti a un processo di interpretazione che li ha resi significativi per il destinatario.

L'informazione è:

- insieme di dati
- interpretati
- comprensibili per il destinatario

DATI: DIAMANTI GREZZI



DATI



INFORMAZIONI

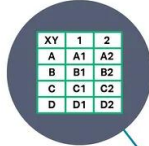


Structured Data

vs

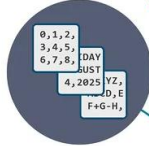
Unstructured Data

Can be displayed
in rows, columns and
relational databases



XY	1	2
A	A1	A2
B	B1	B2
C	C1	C2
D	D1	D2

Numbers, dates
and strings



Estimated 20% of
enterprise data (Gartner)



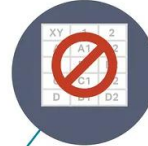
Requires less storage



Easier to manage
and protect with
legacy solutions



Cannot be displayed
in rows, columns and
relational databases



Images, audio, video,
word processing files,
e-mails, spreadsheets



Estimated 80% of
enterprise data (Gartner)



Requires more storage



More difficult to
manage and protect
with legacy solutions



Structured Data

Quantitative - in the form of numbers and values

10%

Unstructured Data

Qualitative - in the form of text files and varying formats

90%

Structured Data

NPS/CSAT	☐
CRM	☐
Sales	☐
Excel	☐
Finance	☐

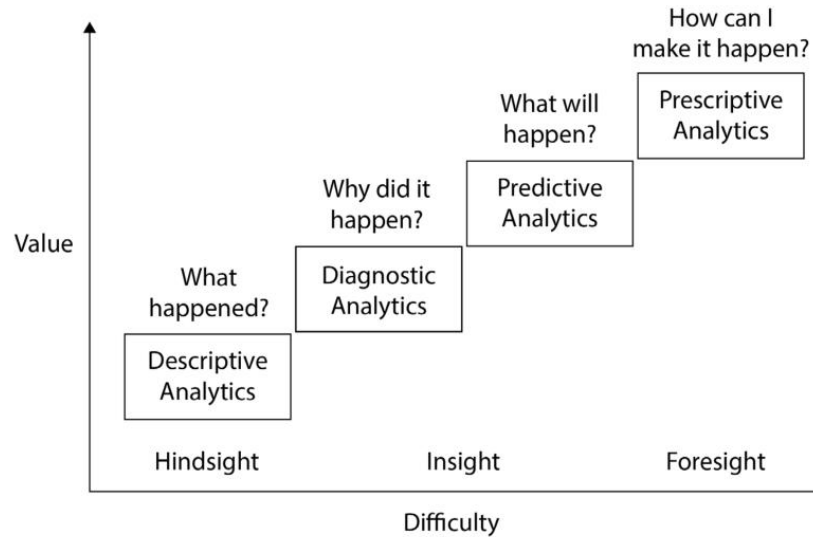
Structured Data

Unstructured Data

Unstructured Data



ANALYTICS PATH TO VALUE (WITH DATA STORY)



Da quanto tempo esiste AI?

The Evolution of A.I.

The 1st. Period 1950-1970

Trivial problem solving, no practicality, GOFAI - Good Old Fashioned Artificial Intelligence

- 1942** The 3 Laws of Robotics by Isaac Asimov. Other sets of laws have been proposed by researchers since then.
- 1950** The Turing Test - proposed by Alan Turing
- 1952** The first self learning game program
- 1956** Dartmouth Conference First Use of the term "Artificial Intelligence/A.I."
- 1957** General Problem Solver (GPS) (by Newell)
- 1958** McCarthy developed LISP programming language
- 1959** The MIT AI Lab (McCarthy and Minsky)
- 1959** The term "Machine Learning" by Samuel
- 1961** First Industrial Robot (Unimate) working at GM
- 1961** SAINT - the first expert system by Slagle (MIT)
- 1964** STUDENT The first A.I. program which understands natural language
- 1965** ELIZA - The first A.I. based chatbot and expert system
- 1966** Shakey - the first locomotive and intelligent robot (SRI)
- 1966** MAC HACK - chess-playing program (by Greenblatt, MIT)

- 1968** SHRDLU, an early natural language understanding computer program
- 1970** WABOT-1 the first anthropomorphic robot (Waseda University)
- 1972** Prolog logic programming language
- 1973** Lighthill Report The poor progress report caused the First A.I. winter (Reduced funding for AI research)
- 1974** MYCIN - The first rule based AI expert system for medical diagnostics
- 1974** The first autonomous vehicle, a mechanical "slider" (Stanford)
- 1980** LISP based machines developed and marketed
- 1980** INTERNIST-I The first Commercial Expert System
- 1986** A driverless van by Mercedes-Benz, with cameras and sensors
- 1988** Bayesian Network (BNs or belief nets) is invented by Pearl
- 1988** The chatbots, Jabberwacky and Cleverbot invented by Carpenter
- 1989** The first autonomous vehicle created by CMU using neural network
- 1993** Polly, the tour guide robot; behavior-based robotics (MIT)

The 2nd. Period 1980-2000

Researchers feeding machines with labeled data. Projects: ICOT - Japan '82, MCC - US '83, Alvey - UK '84. Algorithms began to appear as parts of larger systems. AI solutions proved to be useful throughout the technology industry, such as data mining, industrial robotics, logistics, speech recognition, banking software, medical diagnosis and search engines

- 1997** IBM's Deep Blue beats Gary Kasparov in chess
- 1998** Furby - The first "pet" toy robot for children
- 1999** Kismet - Emotional AI, (MIT AI Lab)
- 1999** AIBO introduced The first AI domestic robot by Sony
- 2000** ASIMO, humanoid robot released by Honda
- 2002** Roomba, autonomous robot vacuum released by i-Robot
- 2004** The first challenge for autonomous vehicles by DARPA
- 2004** NASA rovers Spirit and Opportunity exploring Mars
- 2005** AI based recommendation engines
- 2006** "Machine reading" unsupervised autonomous understanding of text
- 2007** ImageNet - visual database for object recognition software research
- 2007** NVIDIA launches CUDA, a parallel computing platform and programming interface

- 2009** Self Driving Car built by Google. By 2014 it passed Nevada's self-driving test.
- 2009** AI researchers discover GPU (Graphics Processing Unit) for DL
- 2010** Democratize Data Access begins for Image Recognition
- 2010** Narrative Science's AI demonstrates ability to write reports
- 2011** Apple released Siri
- 2011** IBM's Watson wins Jeopardy clash
- 2011** NEIL by CMU, a semantic image analyzer ML system
- 2013** "Vicarious" passes first Turing test - CAPTCHA
- 2014** Microsoft - Cortana
- 2014** Amazon - Alexa
- 2015** Google Brain releases TensorFlow, ML Library, (TPU)
- 2015** Open AI - open source initiative to develop AI benefit of all humanity
- 2016** Google released Google Home
- 2016** Google's DeepMind AlphaGo has defeated Go's No.1 champion
- 2016** NVIDIA announces supercomputer for DL and AI

The 3rd. Period 2010-

The age of machine learning. Computers acquire knowledge from data, not humans. Large tech companies invest in commercial applications of AI/ML.

- 2016** Sophia humanoid robot by Hanson Robotics, the first "robot citizen"
- 2016** PyTorch Open source ML Library
- 2017** The Facebook AI Research Lab trained two chatbots to communicate with each other in order to learn how to negotiate. The chatbots diverged from human language and invented their own language to communicate with one another
- 2017** Caffe Open source DL framework
- 2018** BERT (Google), the first bidirectional, unsupervised language representation
- 2018** Samsung introduces Bixby
- 2018** Facebook detects faces and shares photos with friends to whom those photos belong
- 2018** Alibaba language processing AI outscored human intellect at a Stanford reading and comprehension test
- 2020** DeepMind team uses DL algorithms (Agent57) that outperforms humans at Atari games with deep reinforcement learning
- 2020** Widespread 5G network deployments worldwide

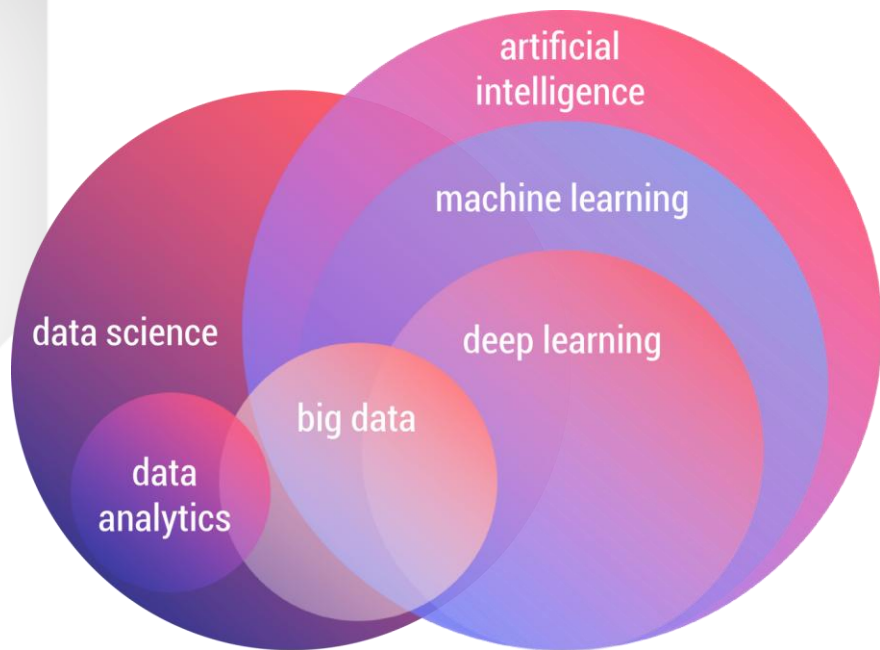
Cos'è AI?

AI: campo della scienza informatica che si occupa dello sviluppo di sistemi e algoritmi in grado di imitare l'intelligenza umana, consentendo alle macchine di apprendere, ragionare, risolvere problemi e prendere decisioni in modo autonomo

L'obiettivo principale dell'AI è quello di creare macchine in grado di imitare e superare le capacità cognitive umane in diversi ambiti

Cos'è il Machine Learning?





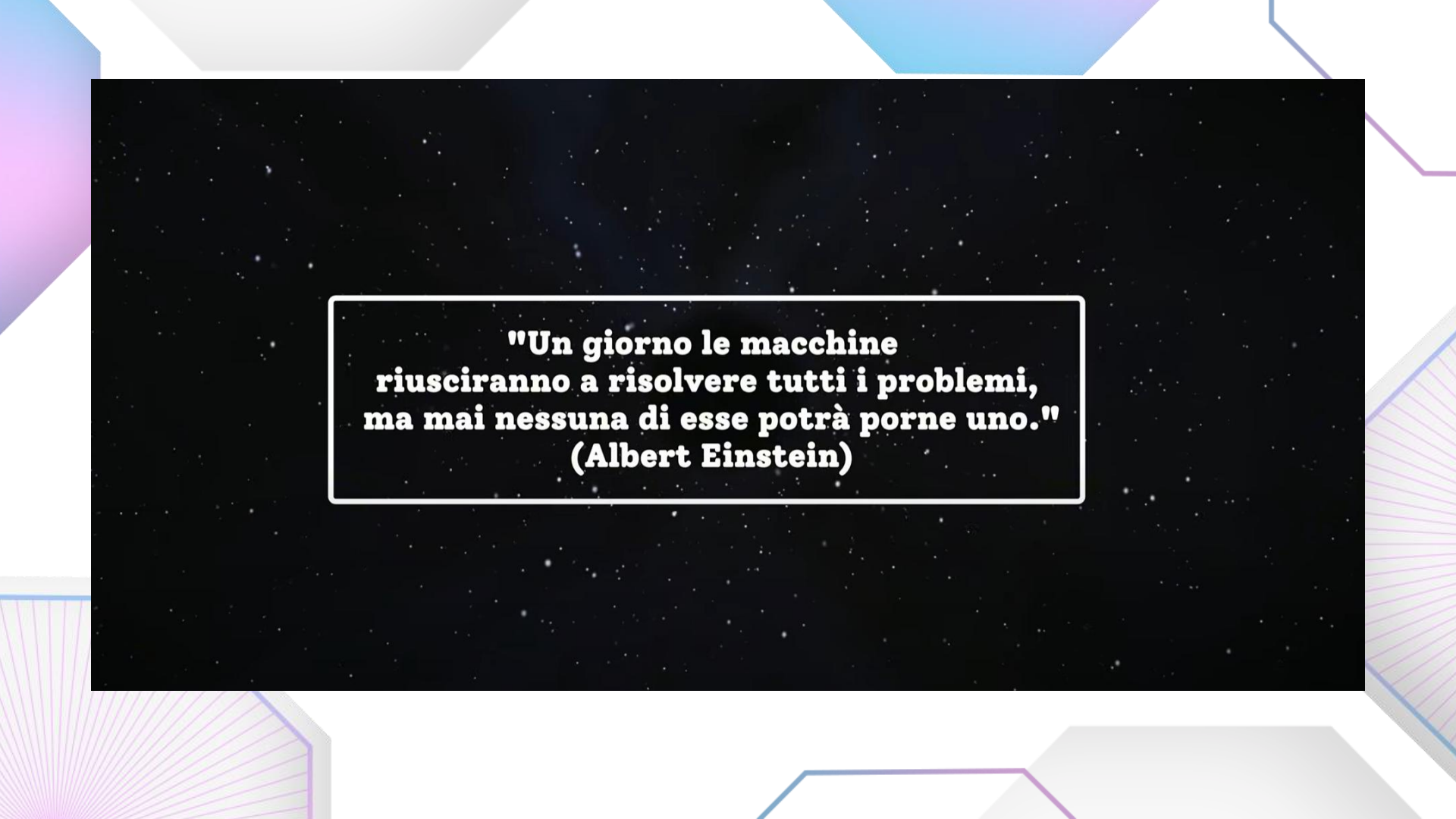
"E' il settore dell'intelligenza artificiale che studia come dare ai computer l'abilità di imparare senza essere esplicitamente programmati" - 1959



ARTHUR SAMUEL

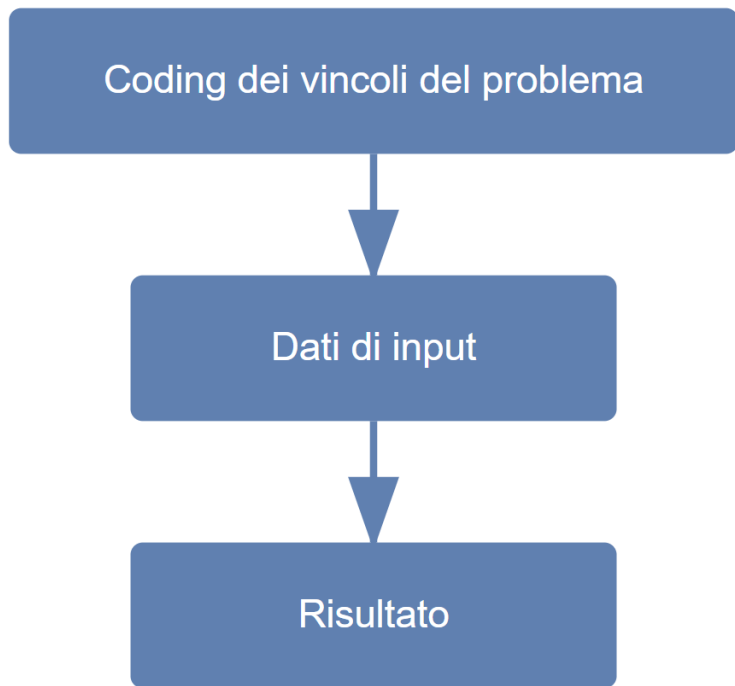
Pioniere dell'AI
Inventore del termine "Machine learning"



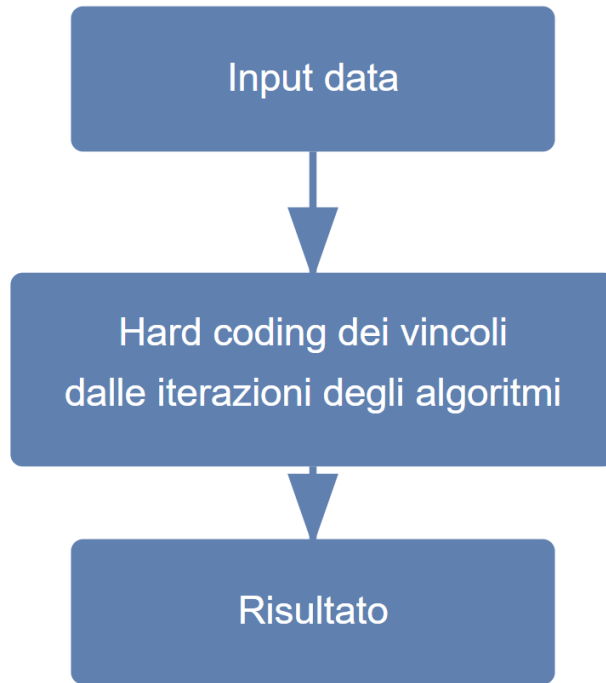


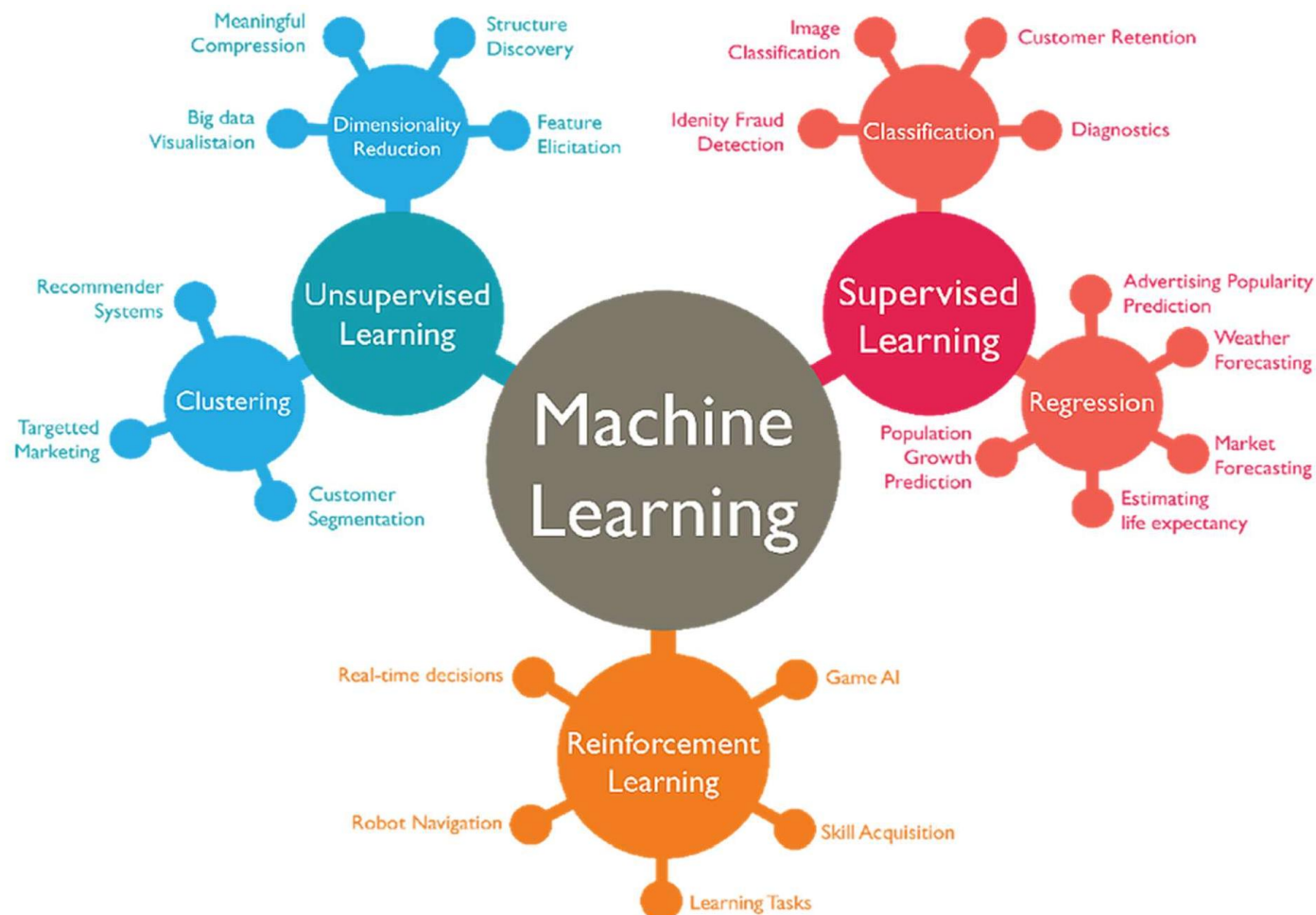
**"Un giorno le macchine
riusciranno a risolvere tutti i problemi,
ma mai nessuna di esse potrà porne uno."
(Albert Einstein)**

Programmazione classica



Machine learning

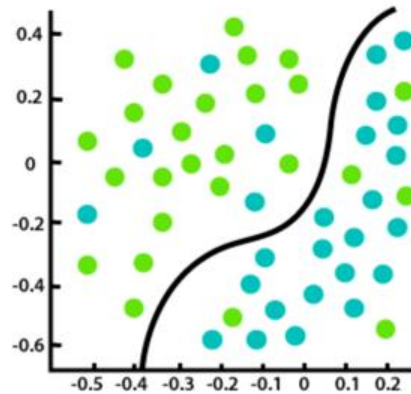




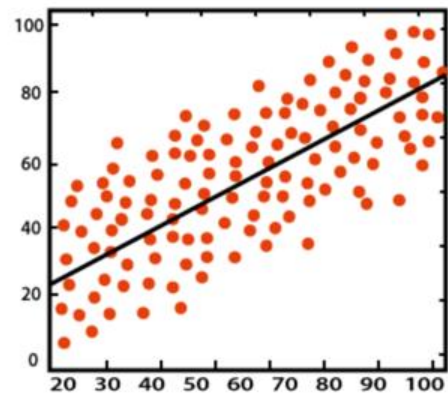
SUPERVISIONATO



Classification



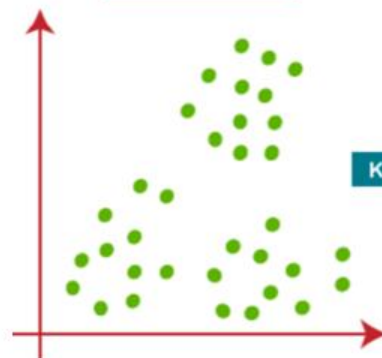
Regression



NON SUPERVISIONATO

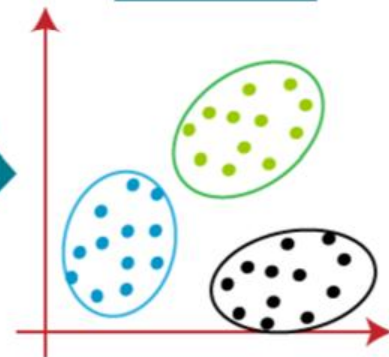


Before K-Means

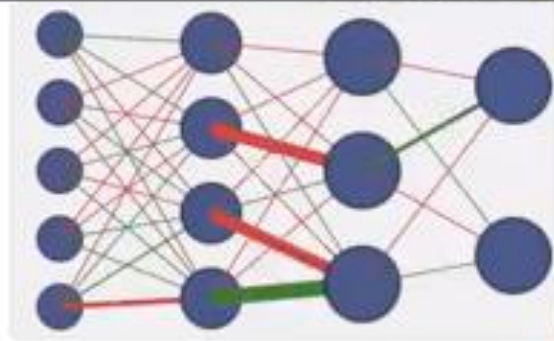
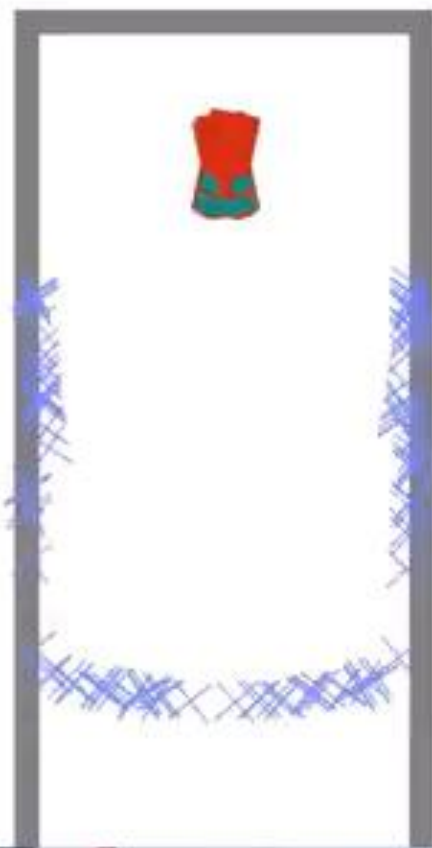


K-Means

After K-Means

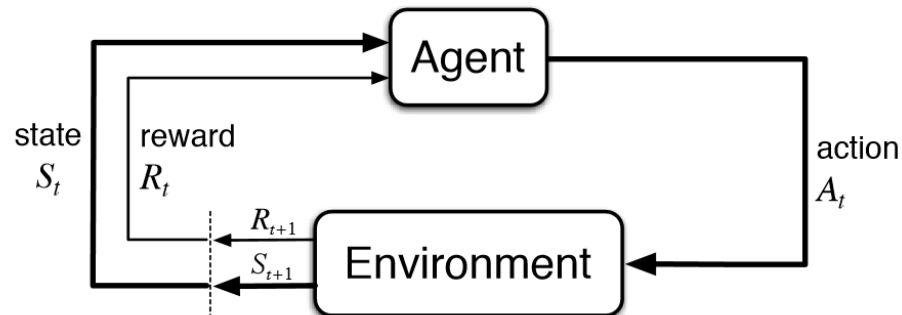


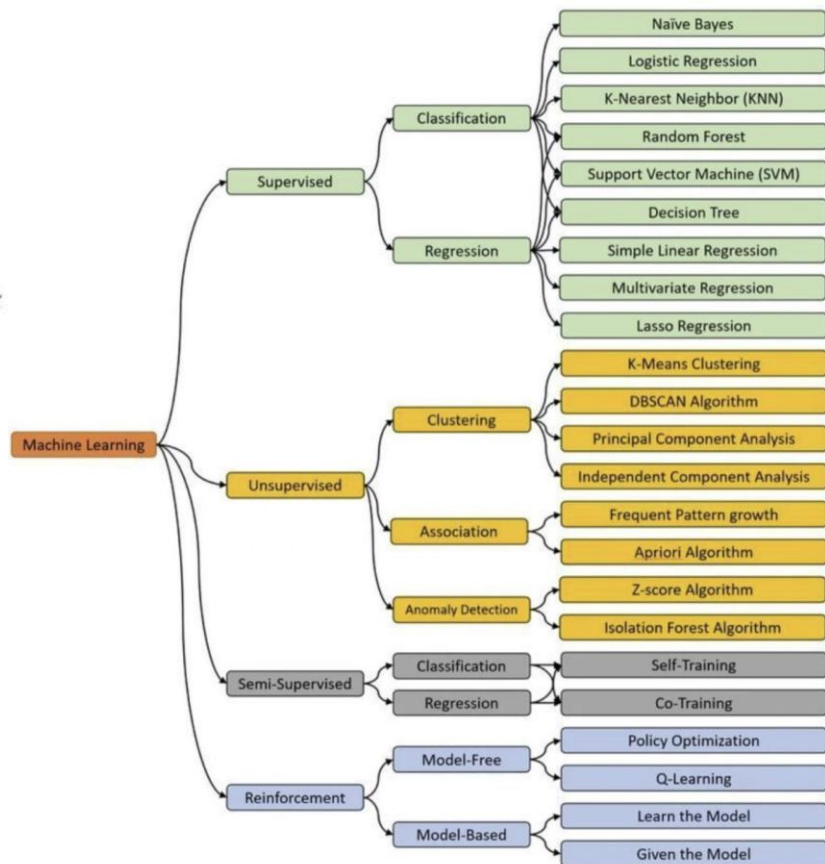
Turn: -0.7932
Engine: 0.99999
Fitness: 9.78920



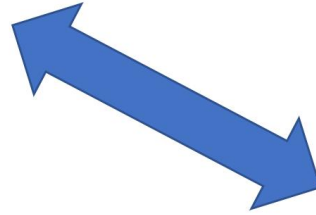
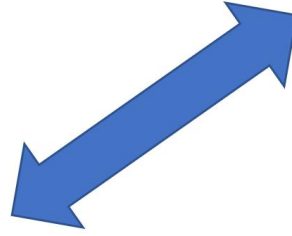
Generation: 1

Reinforcement Learning in ML





AI



+



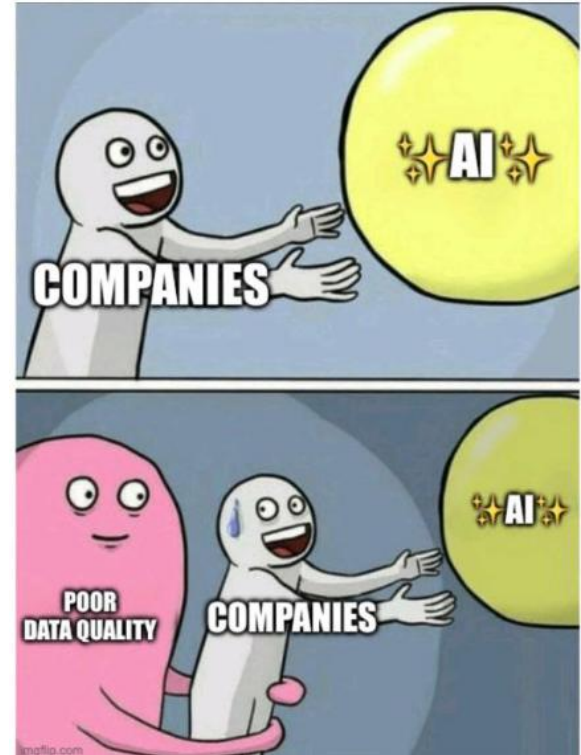
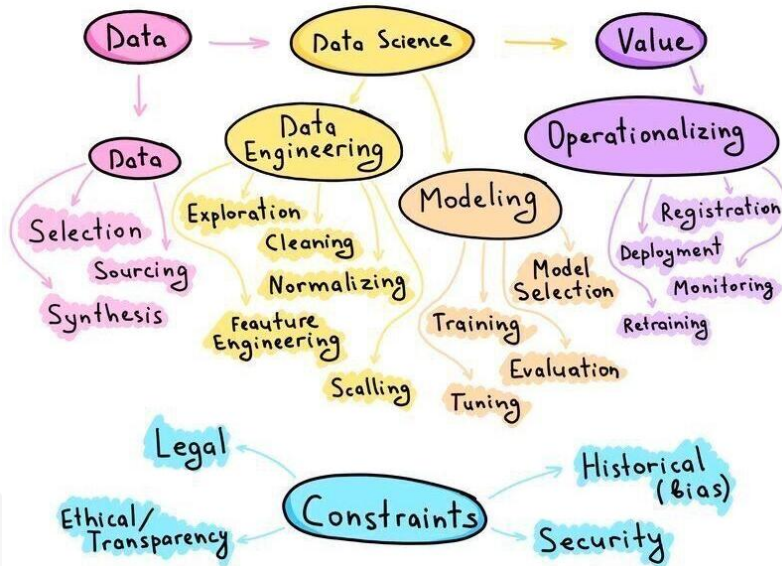
STILL =



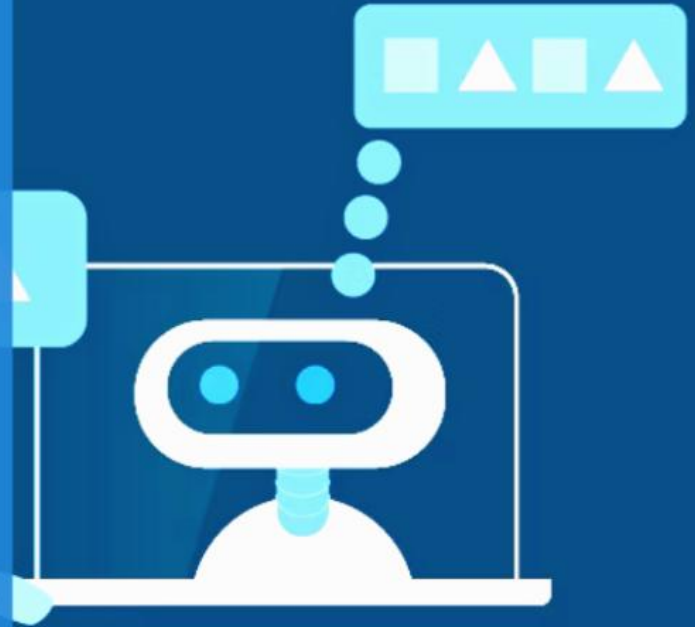
WHAT COMPANIES THINK A.I. LOOKS LIKE

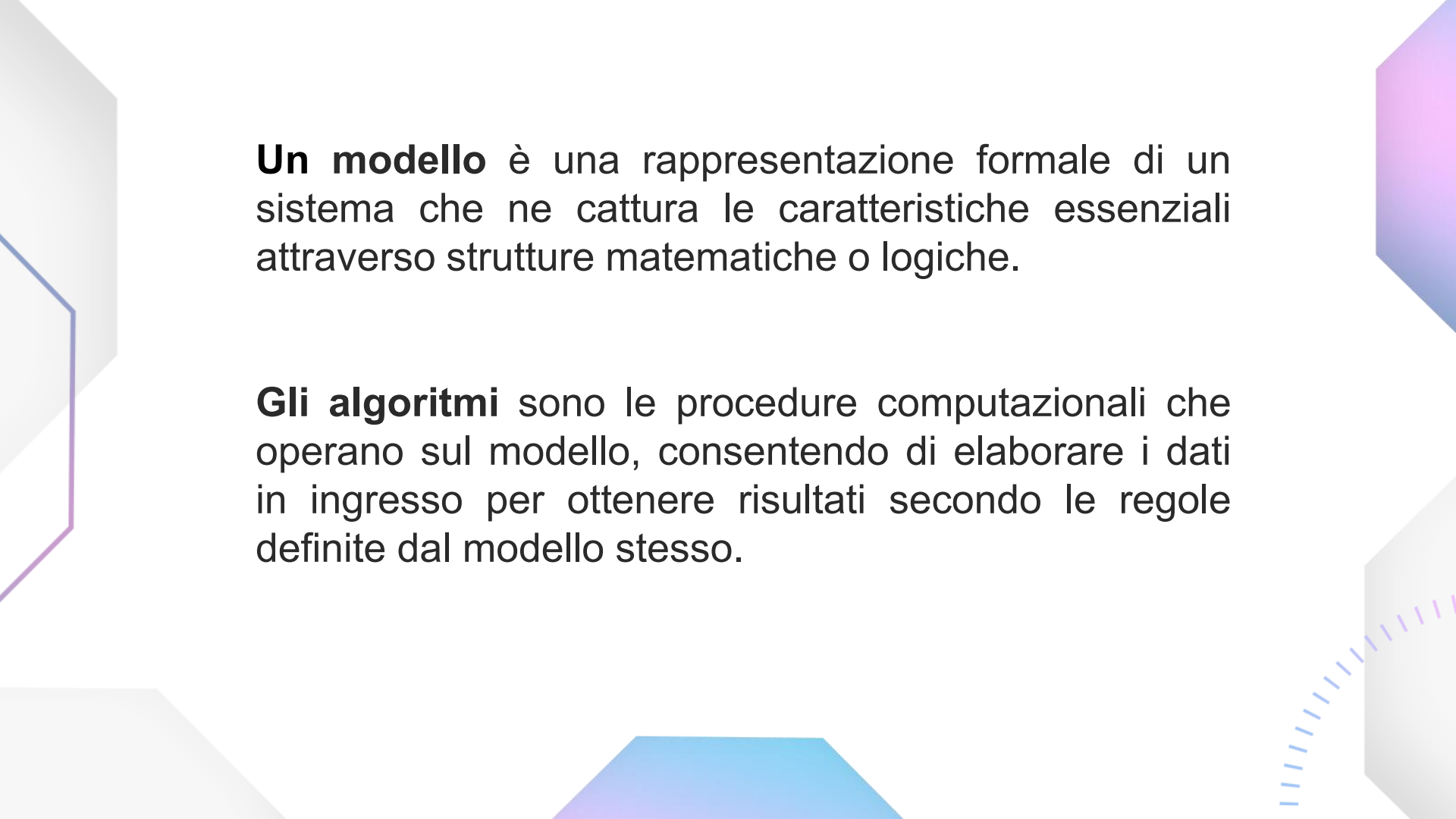


WHAT IT ACTUALLY IS



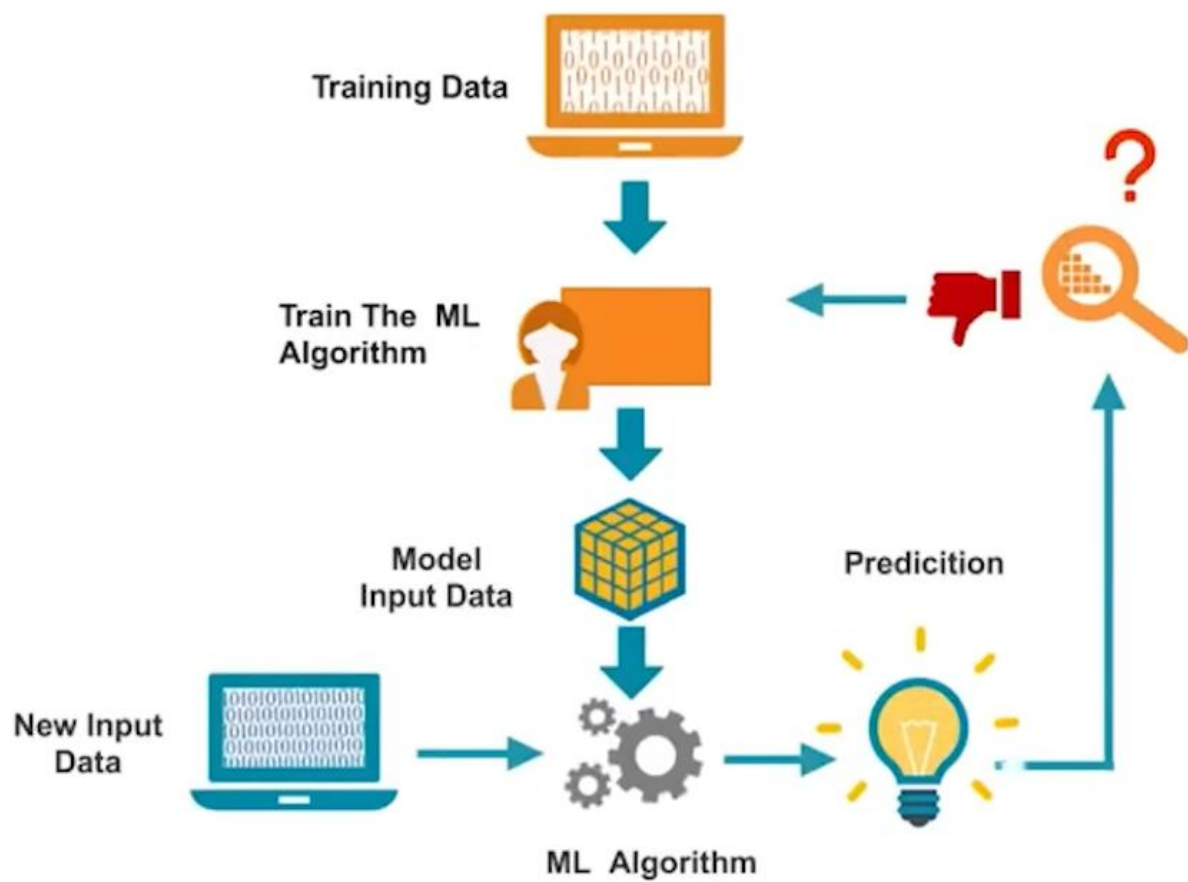
Cos'è un modello di Machine Learning



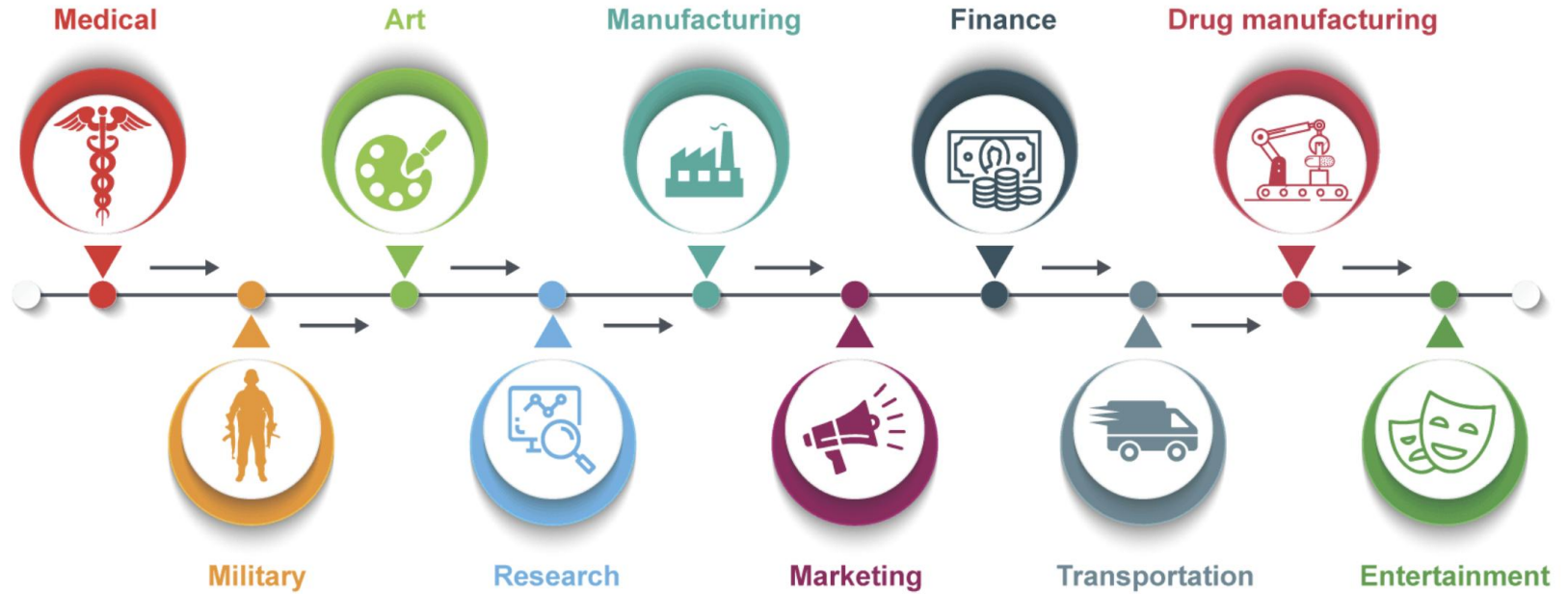
The background features several large, semi-transparent geometric shapes. On the left, there are grey and purple polygons. On the right, there are purple and blue polygons. At the bottom, there are blue and purple polygons. A dashed blue line is visible in the bottom right corner.

Un modello è una rappresentazione formale di un sistema che ne cattura le caratteristiche essenziali attraverso strutture matematiche o logiche.

Gli algoritmi sono le procedure computazionali che operano sul modello, consentendo di elaborare i dati in ingresso per ottenere risultati secondo le regole definite dal modello stesso.

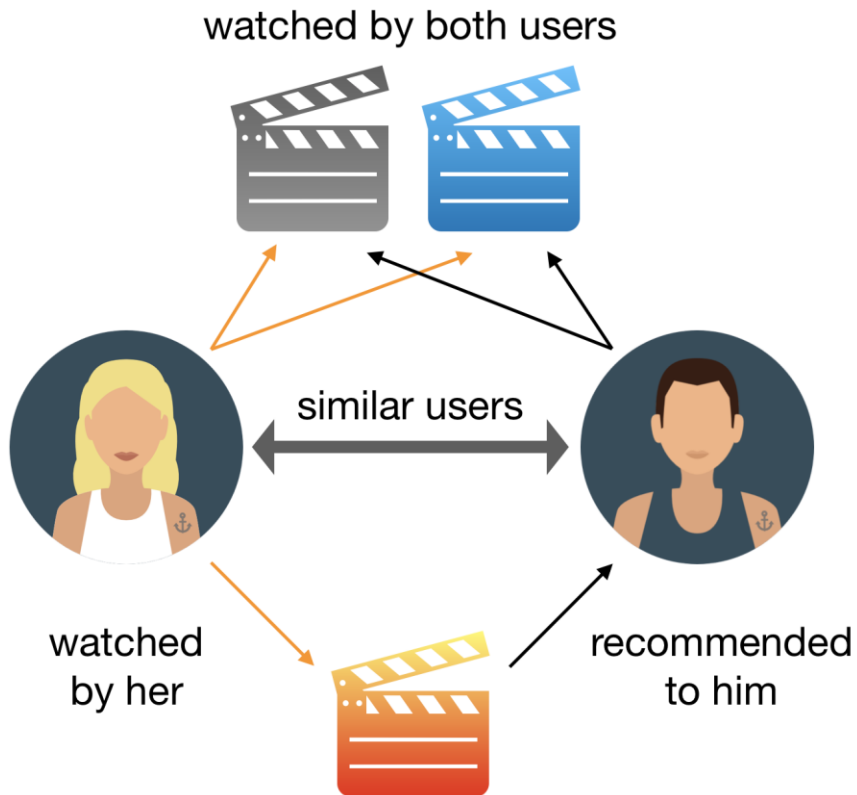


Artificial Intelligence is applicable in a variety of fields such as





RECOMMENDER SYSTEM



Perché solo Oggi?



MAGGIORE POTENZA DI CALCOLO DISPONIBILE

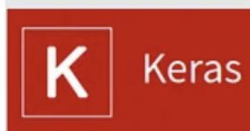


DISPONIBILITA' DI ENORMI QUANTITA' DI DATI

Tool Based Frameworks



Open Source Frameworks



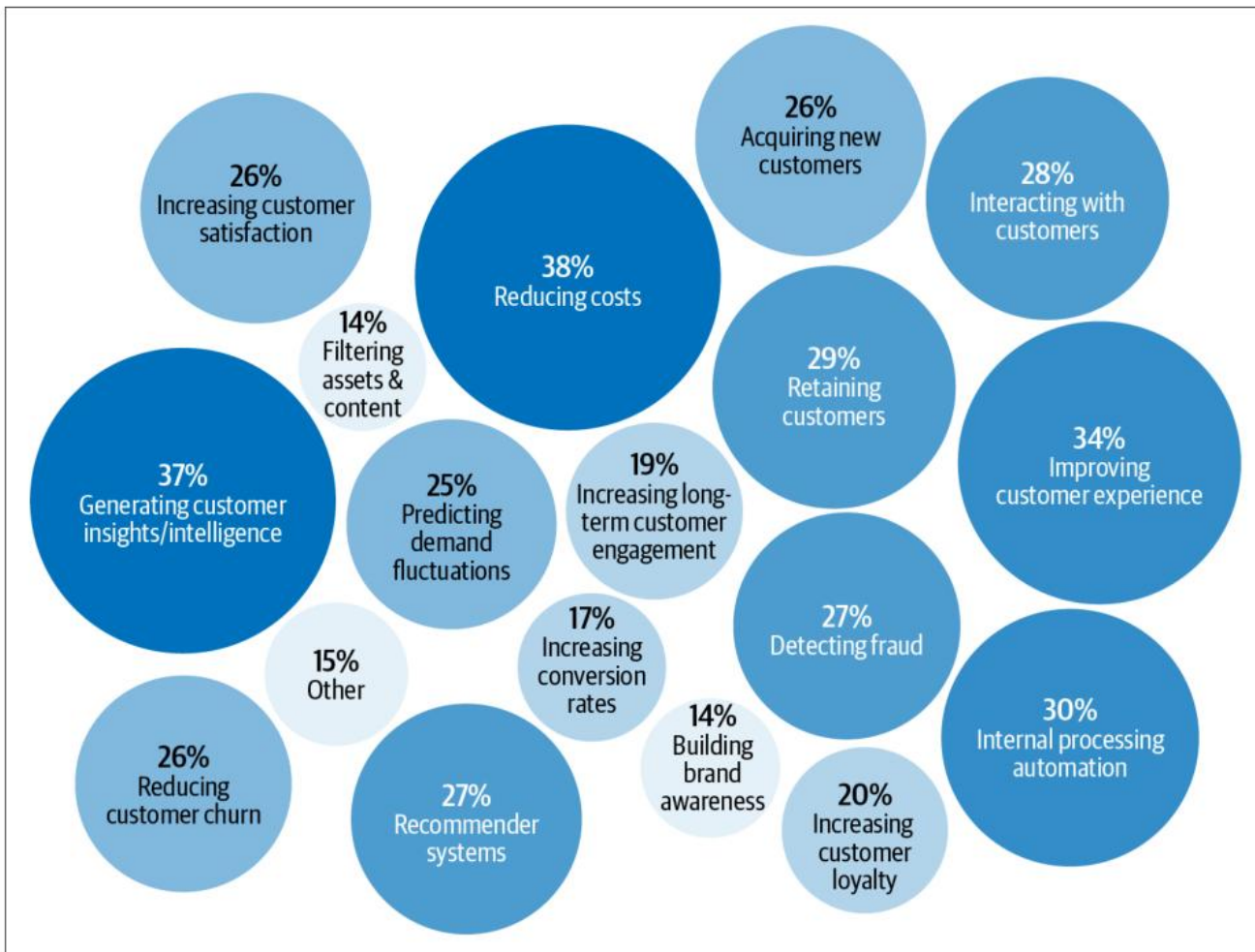
Cloud Based Frameworks



Perché lo si utilizza?

Il machine learning aiuta le aziende a dare un senso ai loro dati, indipendentemente dalla loro dimensione e dal settore di mercato in cui operano.

- Comprendere velocemente le informazioni
- Mostrare i fenomeni che determinano certi andamenti
- Evidenziare i trend emergenti
- Identificare relazioni e pattern nascosti
- Condividere le proprie scoperte con gli altri



Limiti del Machine Learning

Limiti ML Classico su Dati Non Strutturati

IMMAGINI

Non solo pixel ma forme complesse che richiedono filtri convolutivi

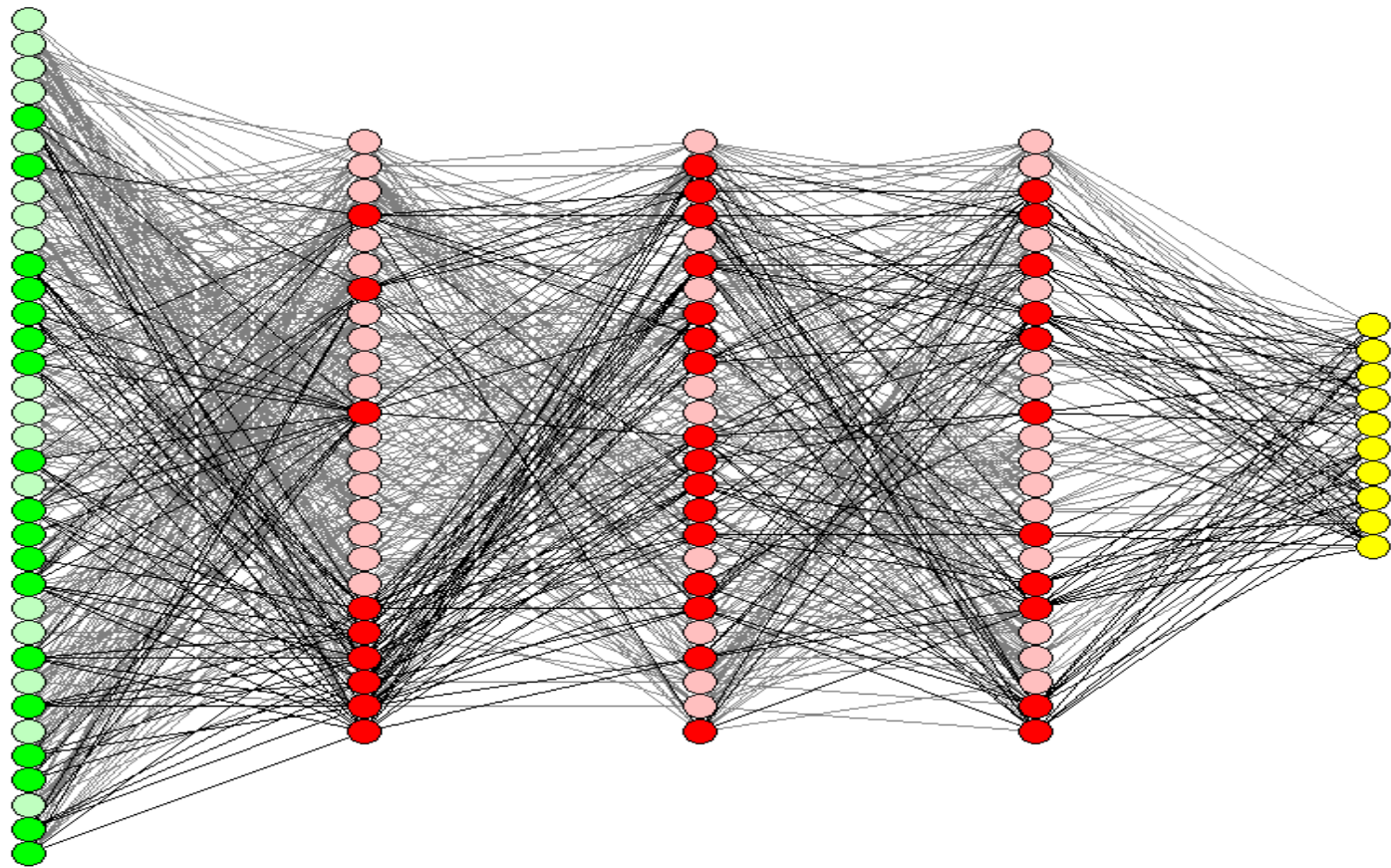
TESTO

Richiede analisi della posizione e del contesto delle parole

ML Classico: Limiti Strutturali

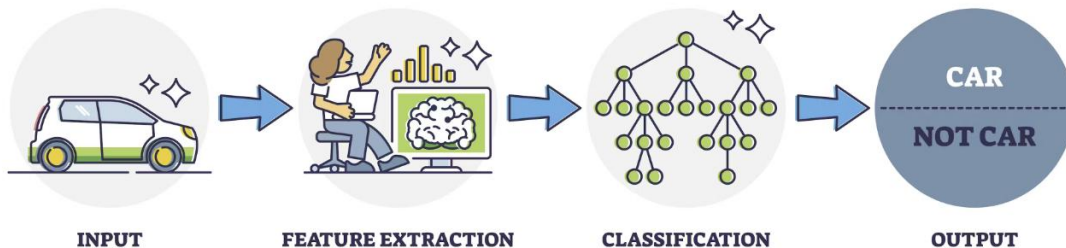
Non può gestire automaticamente le relazioni spaziali e contestuali complesse presenti nei dati non strutturati

DEEP LEARNING

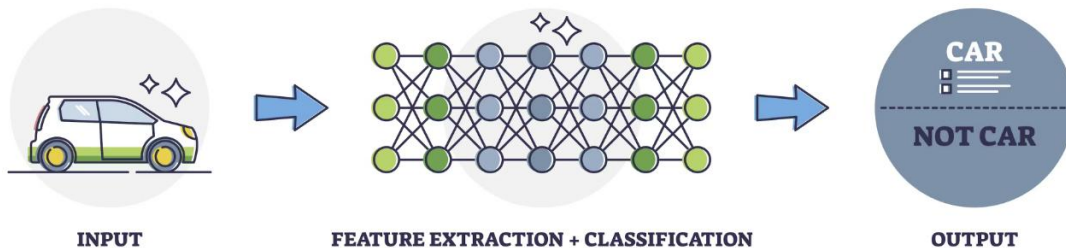


Quali sono i Vantaggi del Deep Learning?

MACHINE LEARNING



DEEP LEARNING



A mostly complete chart of

Neural Networks

©2016 Fjodor van Veen - asimovinstitute.org

- Backfed Input Cell
- Input Cell
- Noisy Input Cell
- Hidden Cell
- Probabilistic Hidden Cell
- Spiking Hidden Cell
- Output Cell
- Match Input Output Cell
- Recurrent Cell
- Memory Cell
- Different Memory Cell
- Kernel
- Convolution or Pool

Perceptron (P)



Feed Forward (FF)



Radial Basis Network (RBF)



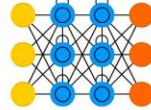
Deep Feed Forward (DFF)



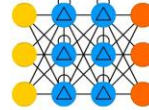
Recurrent Neural Network (RNN)



Long / Short Term Memory (LSTM)



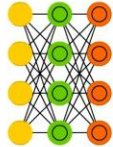
Gated Recurrent Unit (GRU)



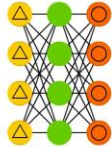
Auto Encoder (AE)



Variational AE (VAE)



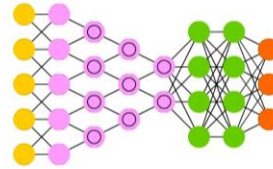
Denosing AE (DAE)



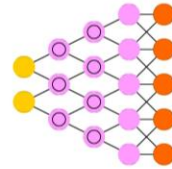
Sparse AE (SAE)



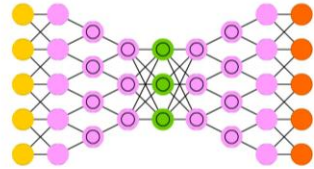
Deep Convolutional Network (DCN)



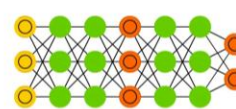
Deconvolutional Network (DN)



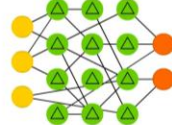
Deep Convolutional Inverse Graphics Network (DCIGN)



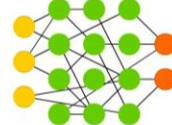
Generative Adversarial Network (GAN)



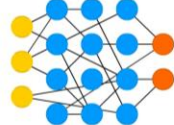
Liquid State Machine (LSM)



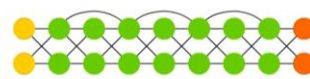
Extreme Learning Machine (ELM)



Echo State Network (ESN)



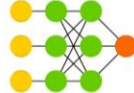
Deep Residual Network (DRN)



Kohonen Network (KN)



Support Vector Machine (SVM)



Neural Turing Machine (NTM)



Markov Chain (MC)



Hopfield Network (HN)



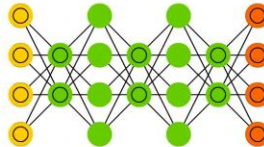
Boltzmann Machine (BM)



Restricted BM (RBM)



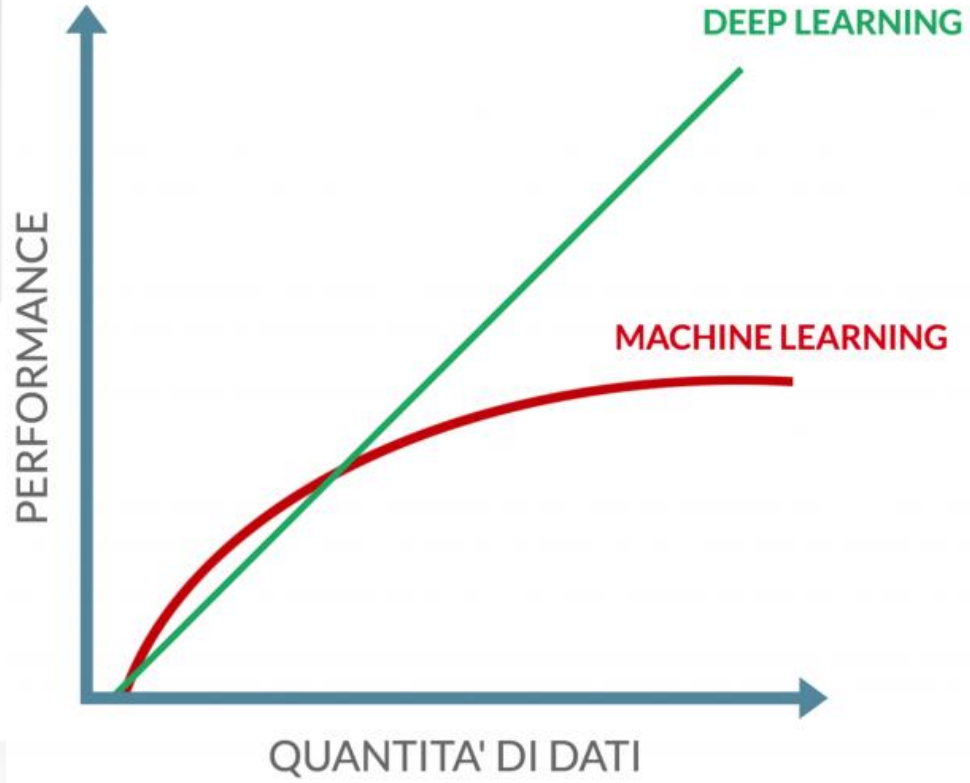
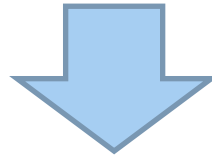
Deep Belief Network (DBN)



<https://setosa.io/ev/image-kernels/>

<https://playground.tensorflow.org/>

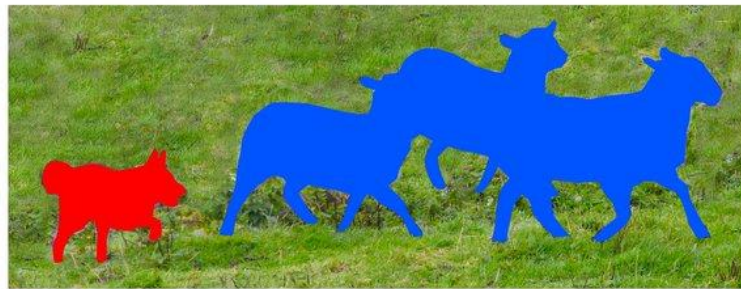
SIMPLE PROBLEM WITH NEURAL NETWORKS



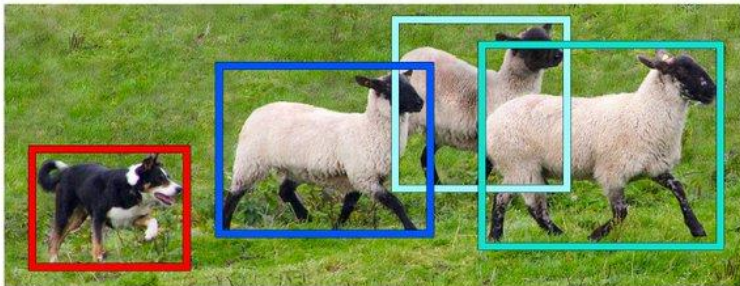
Computer Vision



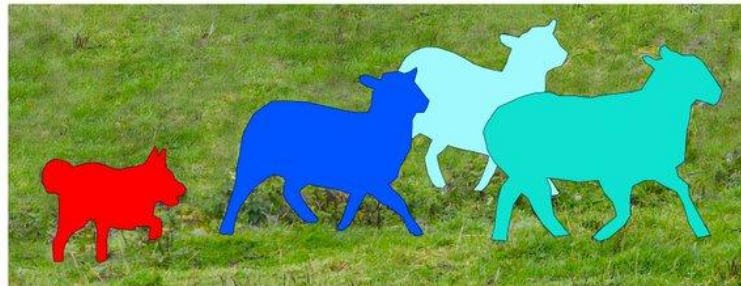
Image Recognition



Semantic Segmentation



Object Detection



Instance Segmentation



Video description



DISCRIMINATIVE vs. GENERATIVE

Discriminative Model

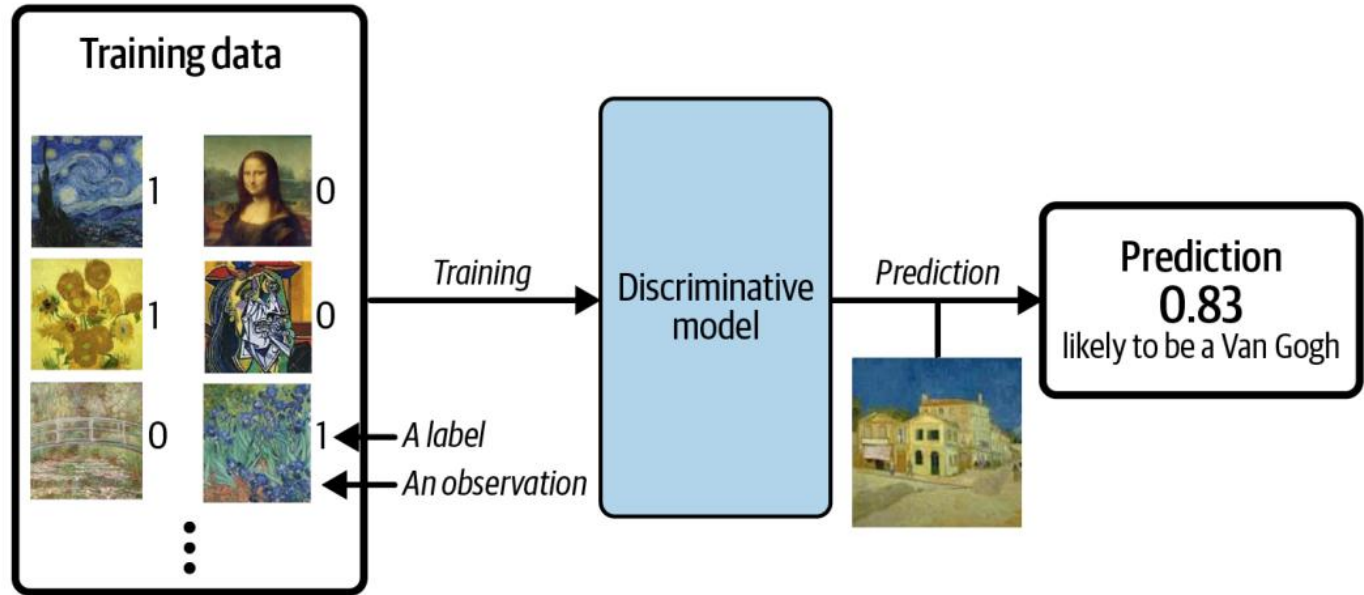


Figure 1-2. A discriminative model trained to predict if a given image is painted by Van Gogh

What Is Generative Modeling?

Generative modeling can be broadly defined as follows:

Generative modeling is a branch of machine learning that involves training a model to produce new data that is similar to a given dataset.

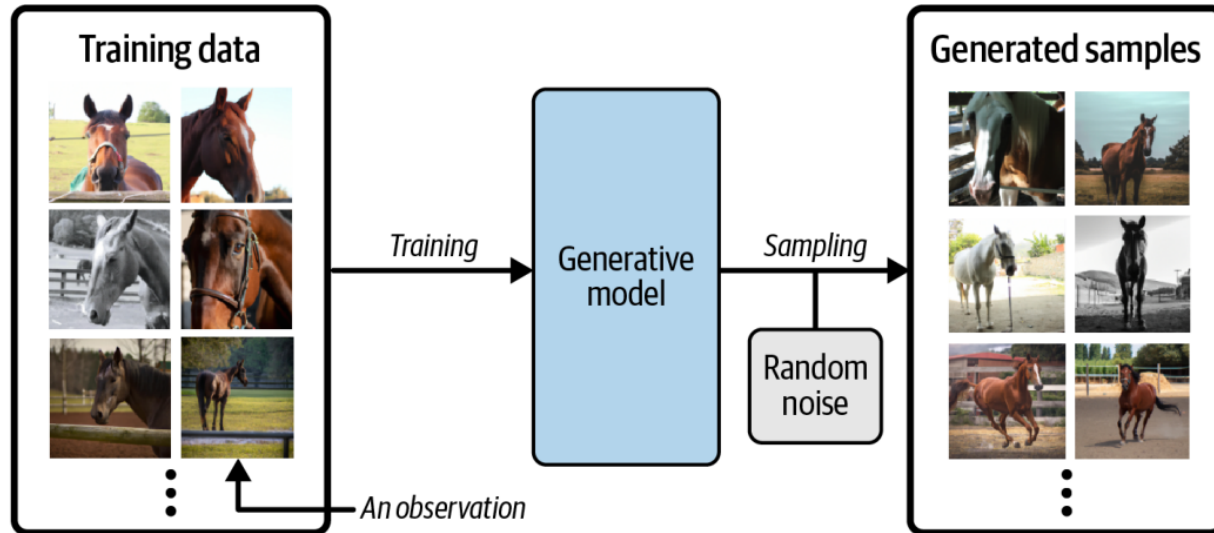
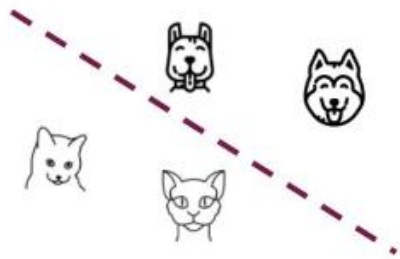


Figure 1-1. A generative model trained to generate realistic photos of horses

Generative Models vs. Discriminative Models

Discriminative models



Features Class

$$X \rightarrow Y$$

$$P(Y|X)$$

Generative models

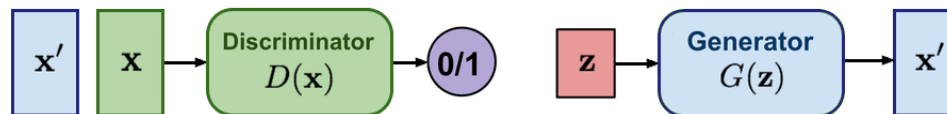


Noise Class Features

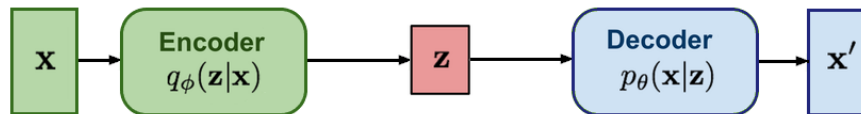
$$\xi, Y \rightarrow X$$

$$P(X|Y)$$

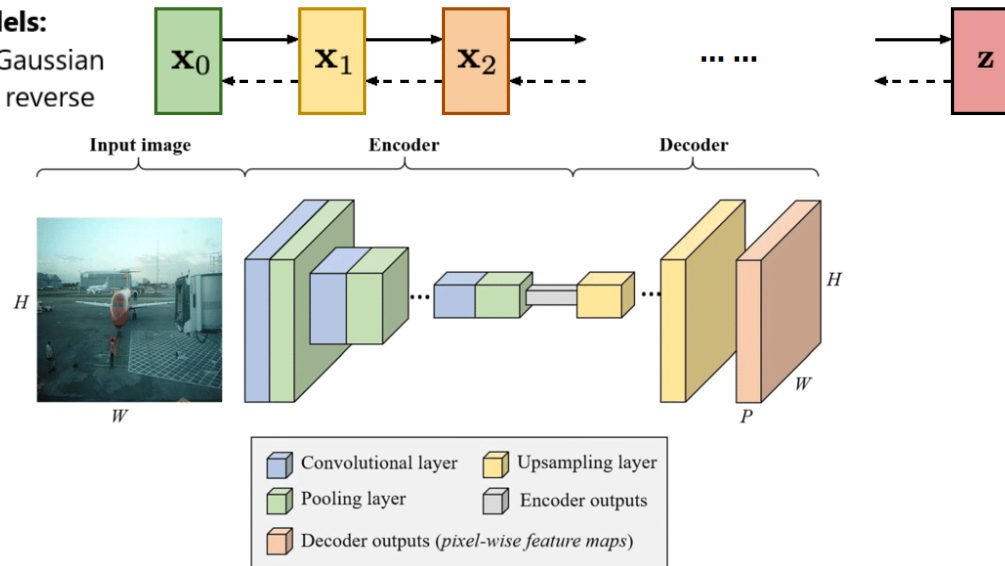
GAN: Adversarial training



VAE: maximize variational lower bound



Diffusion models:
Gradually add Gaussian noise and then reverse





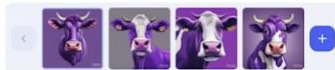
STABLE DIFFUSION XL

SDXL 1.0: A Leap Forward in AI Image Generation

A portrait of a cartoon purple cow, high definition, digital art



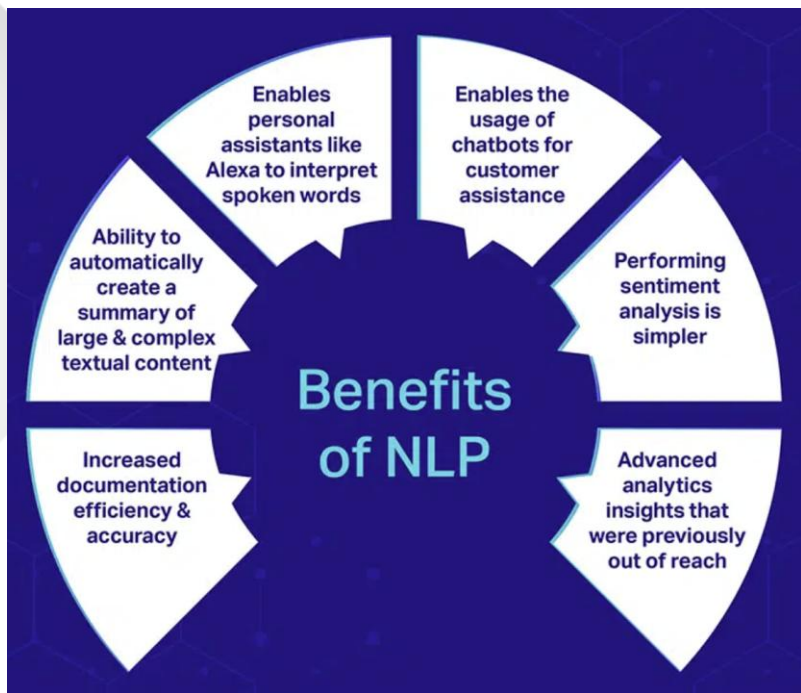
Generate





NLP Market Size & Growth

NLP market is set to reach **\$156.80 billion** by 2030, growing at a CAGR of **27.55%**. Over **85%** of large organizations plan to adopt NLP by 2025, driven by various factors.



Challenges



Misspellings



Language Differences



Innate Biases



Words with
Multiple Meanings



Uncertainty and
False Positives



Training
Data

Application of Text Classification

Sentiment Analysis

Text sentiment, categorizing content as positive, negative, or neutral.

Spam Detection

Identify and filter out spam emails or messages by analyzing their content and characteristics, enhancing communication security.

Topic Labeling

Automatically assigning topics or categories to documents, making content organization and retrieval more efficient.

Language Identification

Detect the language in which a text is written, which is useful for multilingual content processing and translation.

News Categorization

News articles to be categorized into sections like politics, technology, sports, etc., improving content organization for readers.

Product Classification

E-commerce platforms use NLP for product categorization, ensuring items are correctly labeled and presented to customers.

Customer Feedback Analysis

Customer reviews and feedback to extract insights, understand customer satisfaction and areas of improvement.

Medical Document Classification

Categorizing medical records, research papers, and patient notes, assisting in efficient data retrieval for healthcare professionals.

Legal Document Categorization

Law firms use NLP to classify legal documents, making managing and retrieving information from large databases easier.

Social Media Monitoring

NLP tracks and classifies social media posts, tweets, and comments, allowing brands to monitor their online presence and engage with users.

Fraud Detection

Classify financial texts to detect fraudulent activities and identify potential risks.

Resume Screening

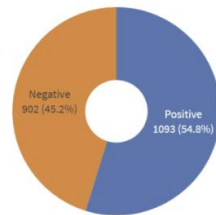
Resume screening by categorizing job applications based on skills, experience, and qualifications.

Content Recommendation

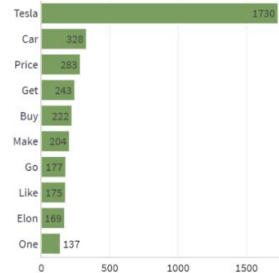
Recommending relevant articles, blogs, or products to users based on their interests.



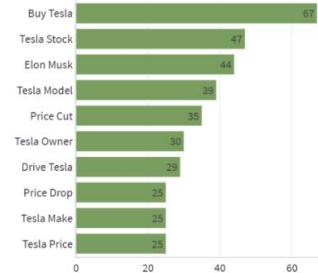
All Positive 😊 Negative 😞



Top 10 Occuring Words



Top 10 Occuring Bigrams



	Sentiment	Tweet
0	Negative	@YatinKheti @sciddoid Again it sounds like some limited release stuff and not re
1	Negative	@3RumsDeep @Tesla @elonmusk @montana_skeptic any bets on if this has an
2	Negative	@ValueAnalyst1 Tesla's demo was obviously hacked together. You can't be that
3	Positive	@rmcadory @farzyness @GM @Toyota @Ford @Tesla Tesla doubling every 12
4	Negative	Well good news from yesterday is the amount I would get for my truck is higher
5	Negative	@alphadoge777 @elonmusk @Tesla @mayemusm Exactly.
6	Positive	I don't like asking for many favors. But this is a time where I am going to need y
7	Negative	@farzyness Redo: you need to compare global to global, Tesla sells 1.3m globa
8	Positive	@stevenbward @Tesla Sweet feelings:) Such a lovely pic 🌸

Wordcloud





Use Cases



Intelligent Doc.
Processing



Sentiment
Analysis



Fraud
Detection



Language
Detection



Conversational
AI / Chatbot



Text
Summarization



Text
Translation



Question-
Answering



Data
Redaction



Social Media
Monitoring

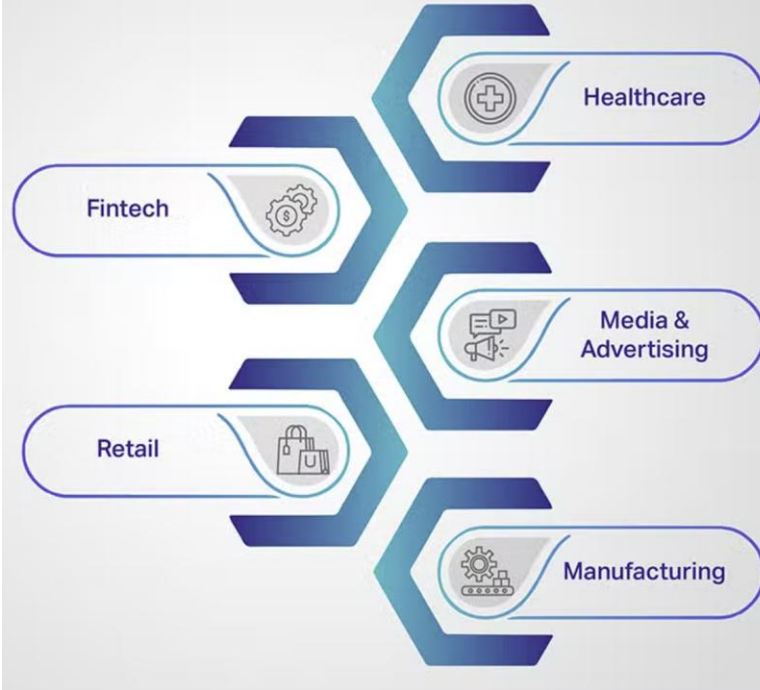


Business
Analytics



Clinical
Documentation

Industries Leveraging NLP



Healthcare

NLP offers rewarding benefits to the healthcare industry such as:

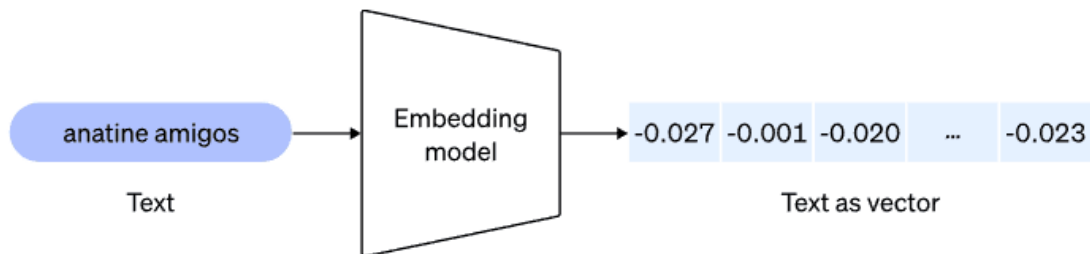
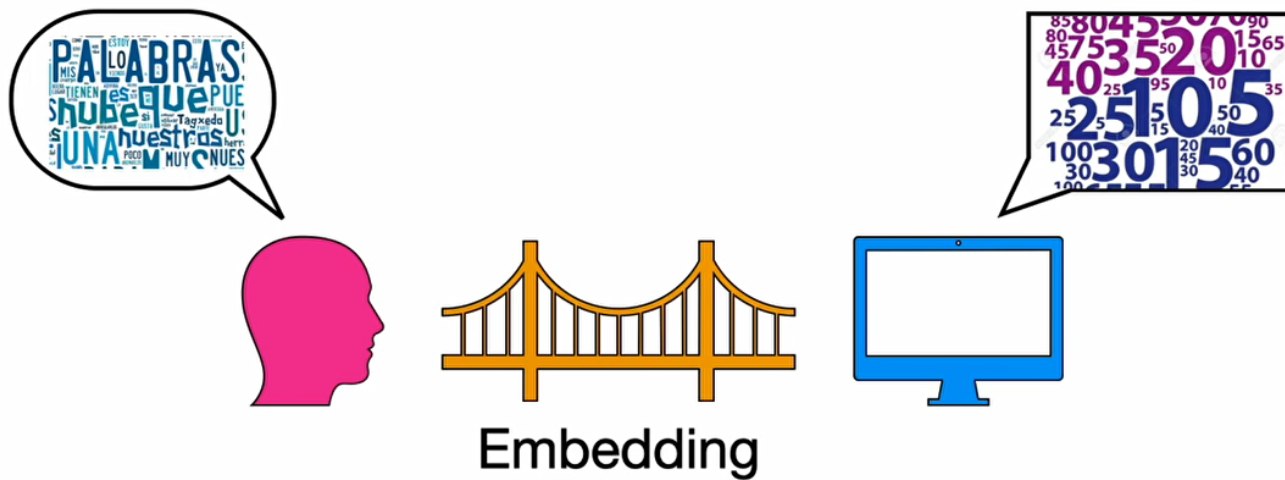
- the extraction insights from medical records and analysis of unstructured data
- Improve and personalize clinical decision support systems
- Optimize responses from chatbots for seamless patient care experiences
- Monitor, predict, and mitigate adverse drug reactions and implement pharmacovigilance strategies and more

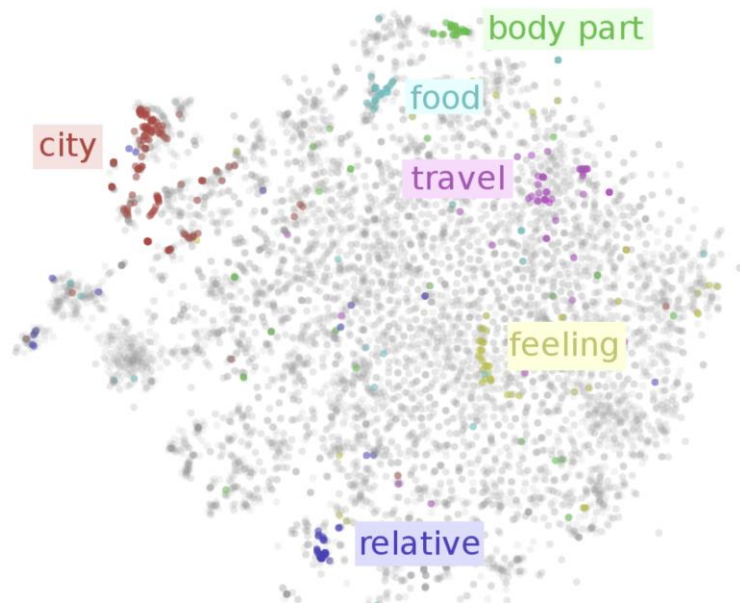
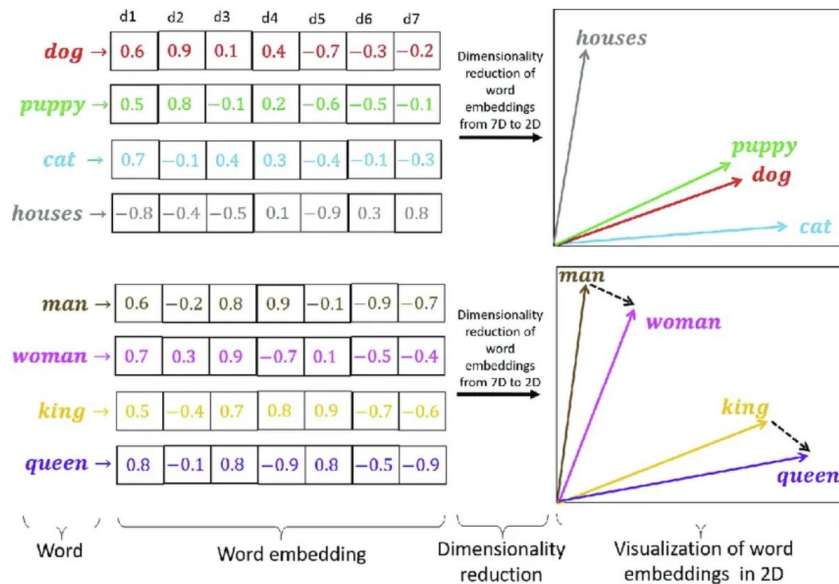
Fintech

The implications of NLP in fintech is completely different, offering benefits like:

- Seamless document processing and onboarding
- Optimize risk management and fraud detection
- Assessment of creditworthiness of individuals for financing
- Personalization of financial products in terms of tenures and premiums and more

Embeddings





<https://projector.tensorflow.org/>

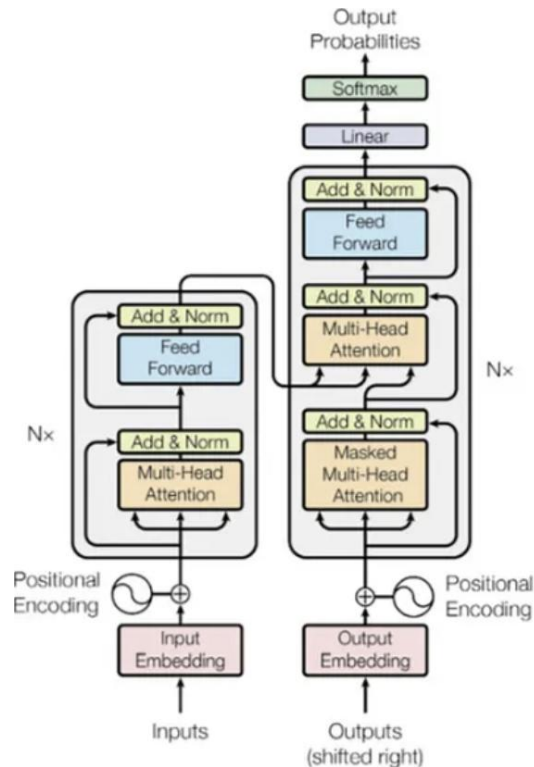
6 Dec 2017

Attention Is All You Need

Jakob Uszkoreit
Google Research
usz@google.com

Lukasz Kaiser*
Google Brain
lukaszkaizer@google.com

The diagram illustrates the difference between a sentence and a paragraph. It shows two sentences, Sentence 1 and Sentence 2, with their constituent words and the connections between them. Sentence 1 is "The animal didn't cross the street because it was too long." Sentence 2 is "The animal didn't cross the street because it was too tired." The words "animal", "street", and "it" are highlighted in orange in both sentences, indicating they are shared across the two sentences. The connections between words are shown by lines, with some lines being orange to highlight the shared words.



BERT

Google

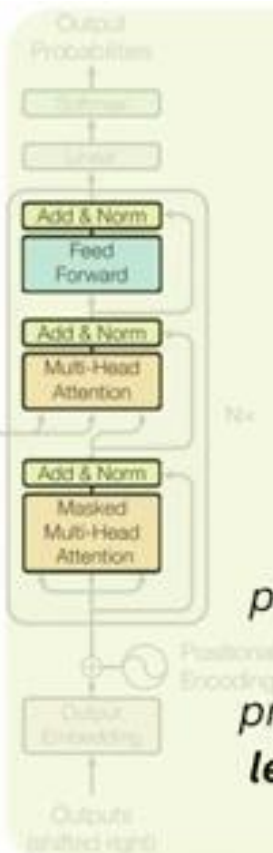
*use transfer learning to **continue learning** from its existing data when adding user-specific tasks and layers.*



GPT

OpenAI

*decodes from its massive pre-learned embeddings to present output that matches user prompts. It does **not learn** anything new.*



LLM

LARGE LANGUAGE MODEL

Text Training Data

Web Data



Books



Images
converted to text



Structural Data



Training

Large
Language Model

Adaption

Adaptations

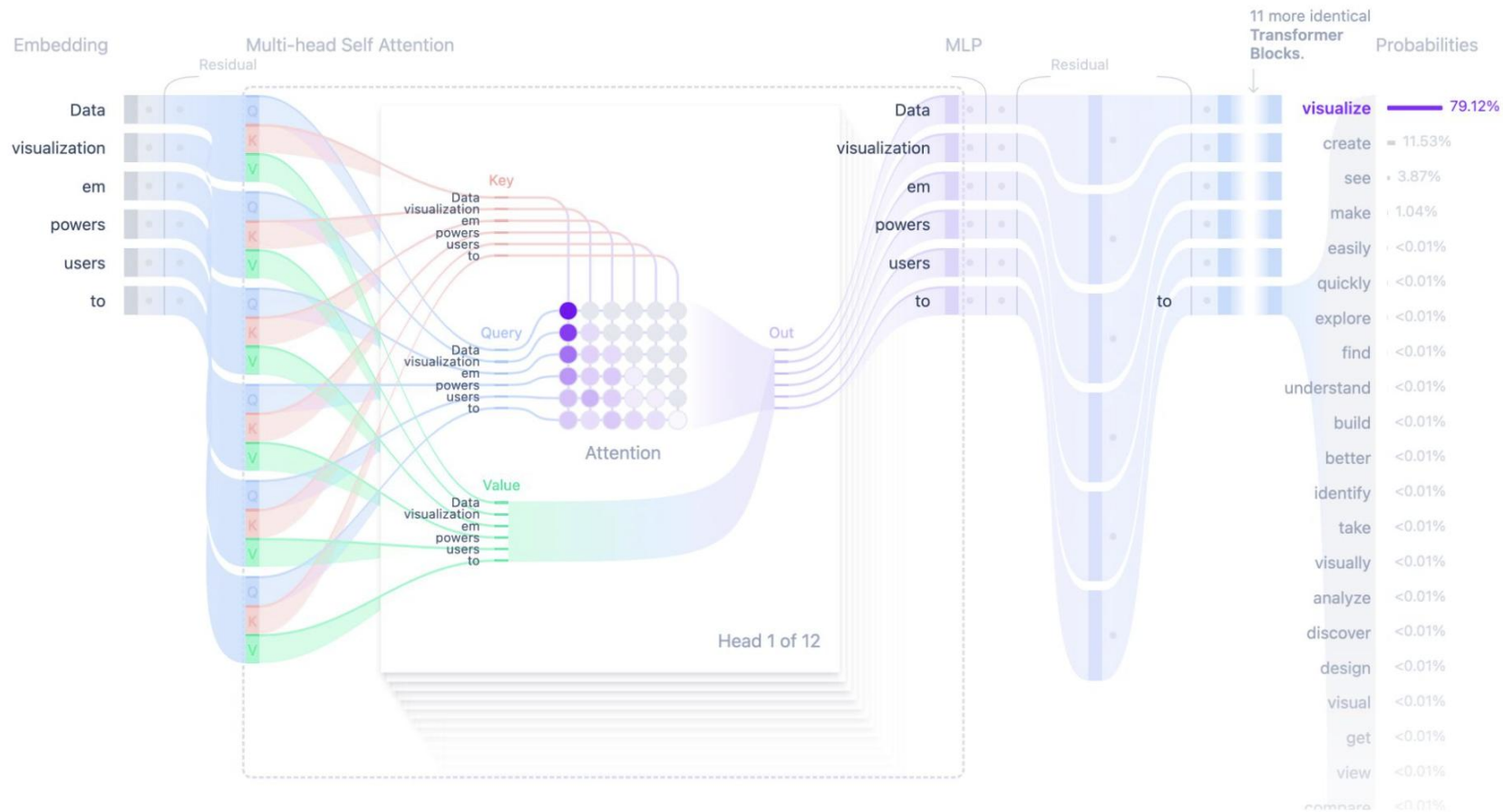
Question
Answering

Sentient
Analysis

Information
Summarization

Planning

Language
Translation



PREFIX

Language Model

NEXT TOKEN

I saw a cat on a

"a"

Hash
Function

Seed

Reed List
Green List

mat	25%
carpet	13% ★
boat	5%
plane	3%
....	

The sky is



LLM

blue = -0.96
clear = -1.60
usually = -2.47
the = -3.40
< = -3.47

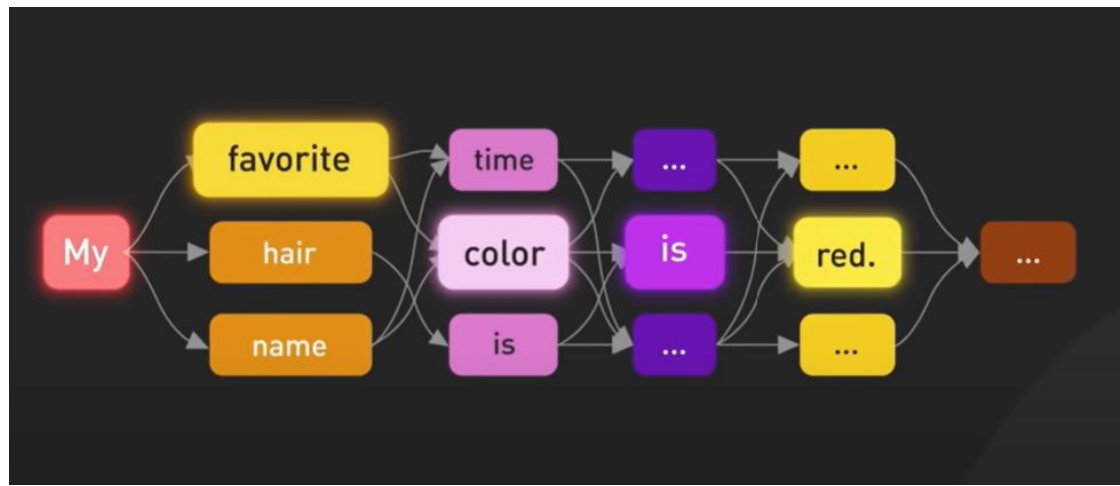
More likely

Less likely

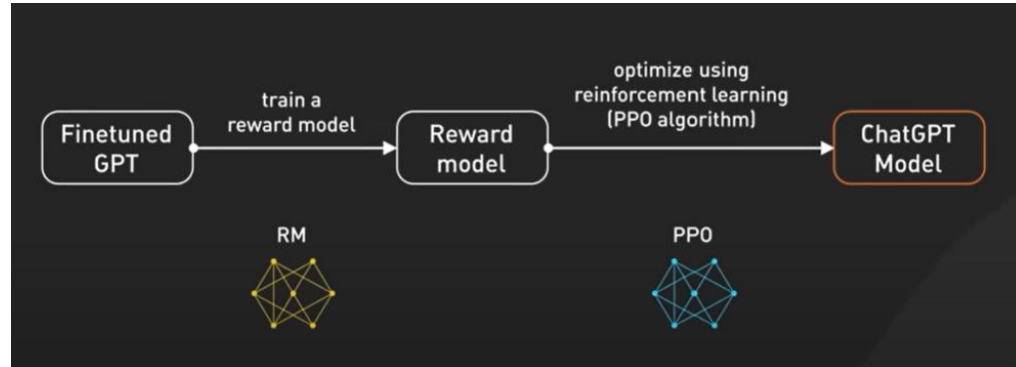
The sky is blue

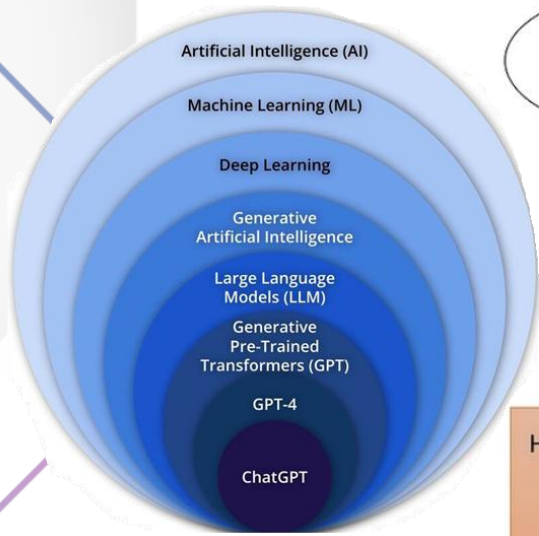
Total: -0.96 logprob on 1 token
(73.18% probability covered in top 5 logits)





Reinforcement Learning with Human Feedback





AI Language Model developed by OpenAI.



Generative

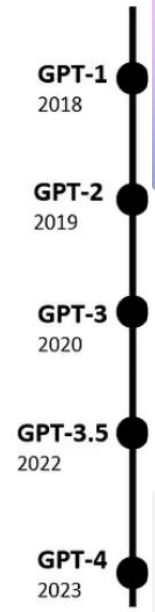
Pre-trained

Transformer

Has creativity to generate original text as of it was written by real person

Is trained with vast amount of text from books, articles, websites, and more.

Architecture that helps it to understand the connections between words, context, and even emotions.



Generative AI

Computers have gotten smarter

Execute Instructions

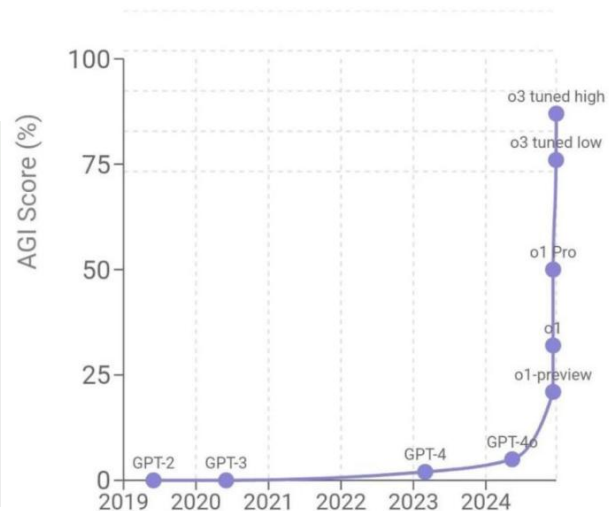


Learn + Think



```
function withdraw(amount) {  
  if (amount > balance) {  
    fail("Hey you ain't got the cash!")  
  } else {  
    balance = balance - amount  
  }  
}
```

ARC AGI Score Progression





You

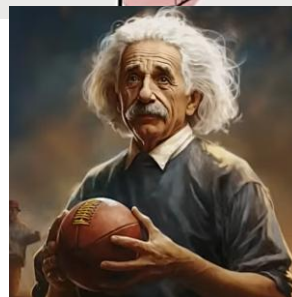
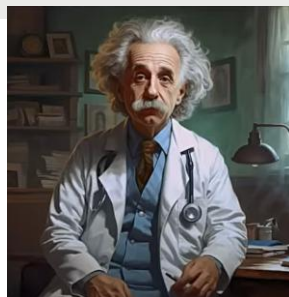
What are you? Explain in 1 sentence to a 10 year old.



ChatGPT

I'm like a super smart robot that can answer lots of questions and help with all sorts of things, kind of like a really helpful genie in a computer!

Einstein in
your basement

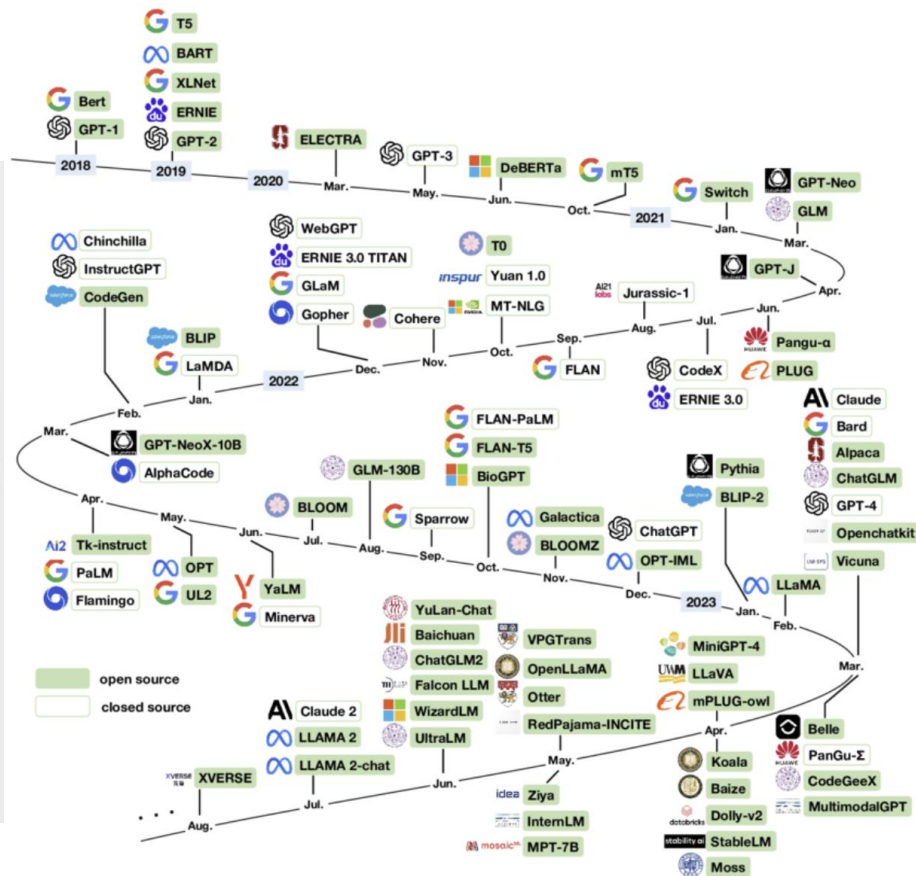


Limiti umani:

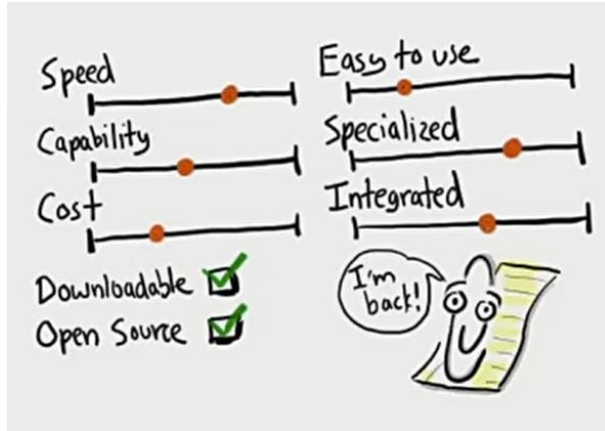
- Fare errori
- Giungere a conclusioni affrettate
- Può fraintendere
- Inventarsi soluzioni che non esistono (allucinazioni)



Models, models everywhere



Open-Source Vs. Close-Source LLMs



Features



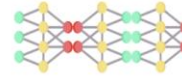
Cost

Innovation

Support

Customization

Open-Source LLMs



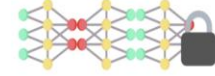
Lower initial cost but possibly higher maintenance.

Collaborative and fast innovation from a global community.

Reliant on community-based support, which can be extensive but inconsistent.

Highly customizable with access to source code.

Close-Source LLMs

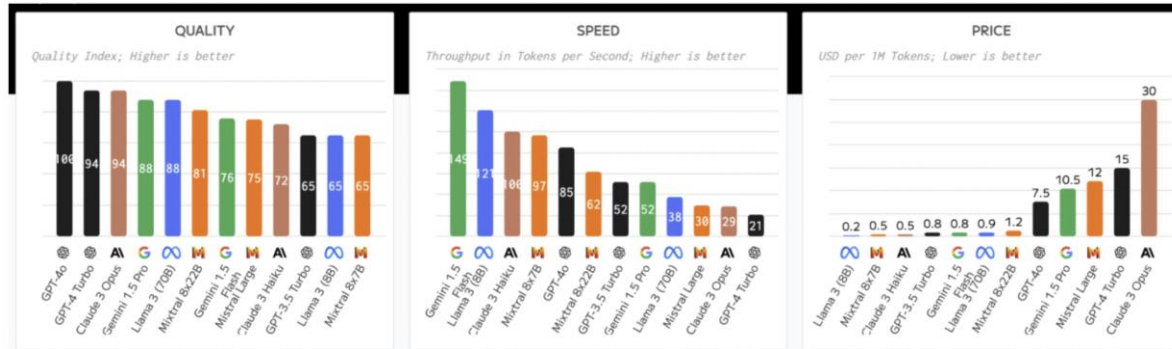


Higher initial cost but lower maintenance.

Slower innovation directed by a single entity.

Consistent and quality support offered through dedicated channels.

Customization possible within the platform's limits.



Model Types

Text to Text

Text to Image

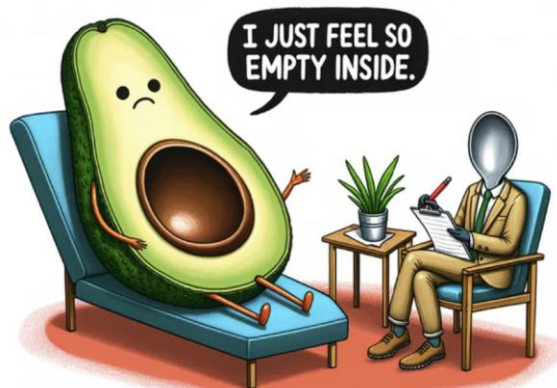
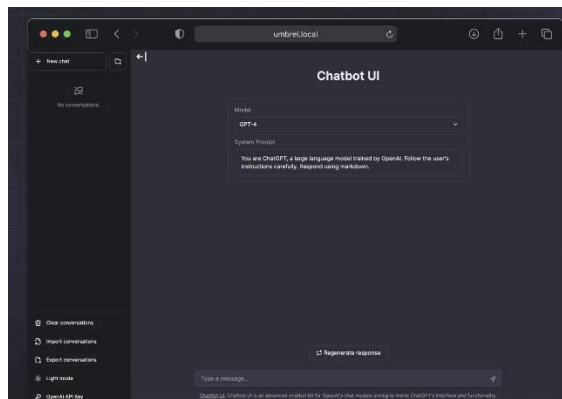
CHAT GPT 3.5



OpenAI

DALL-E 3

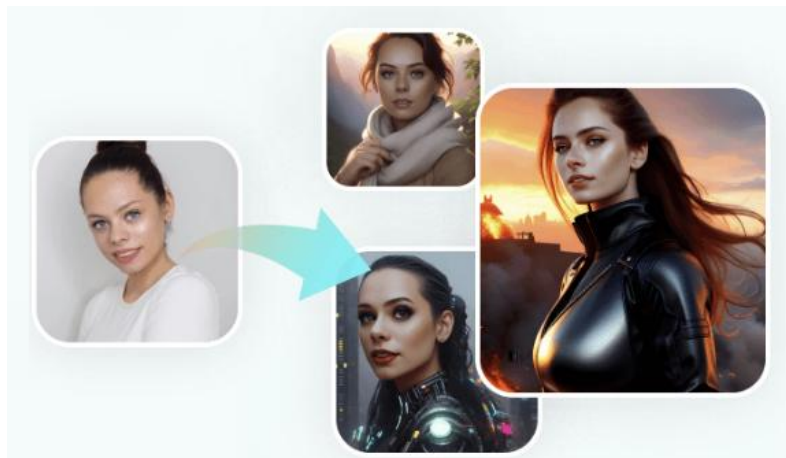
DALL-E 3 understands significantly more nuance and detail than our previous systems, allowing you to easily translate your ideas into exceptionally accurate images.



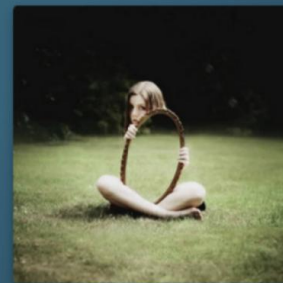
Model Types

Image to Image

Image to Text



What are we looking at here?

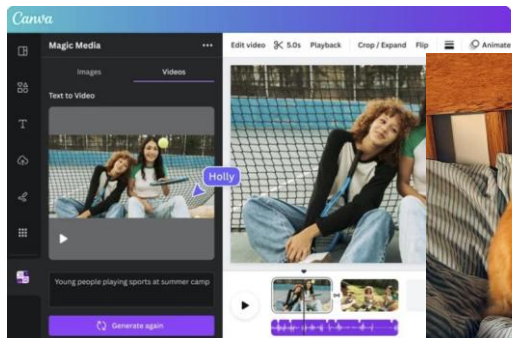


✓ Analyzing the image: Privacy blur hides faces from Bing chat

The image you're seeing is quite intriguing. It features a person sitting cross-legged on a grassy field, holding a large, oval-shaped mirror with a gold frame. The person's face is blurred out, maintaining their privacy. The mirror reflects the surrounding environment, adding an interesting perspective to the scene. The backdrop of the image is a grassy field with trees in the distance, contributing to the overall serene and tranquil atmosphere. The image has a dark and moody tone, which adds a layer of depth and emotion to it. It's a beautiful blend of nature and human presence, don't you think?

Model Types

Text to Video



OpenAI - Sora

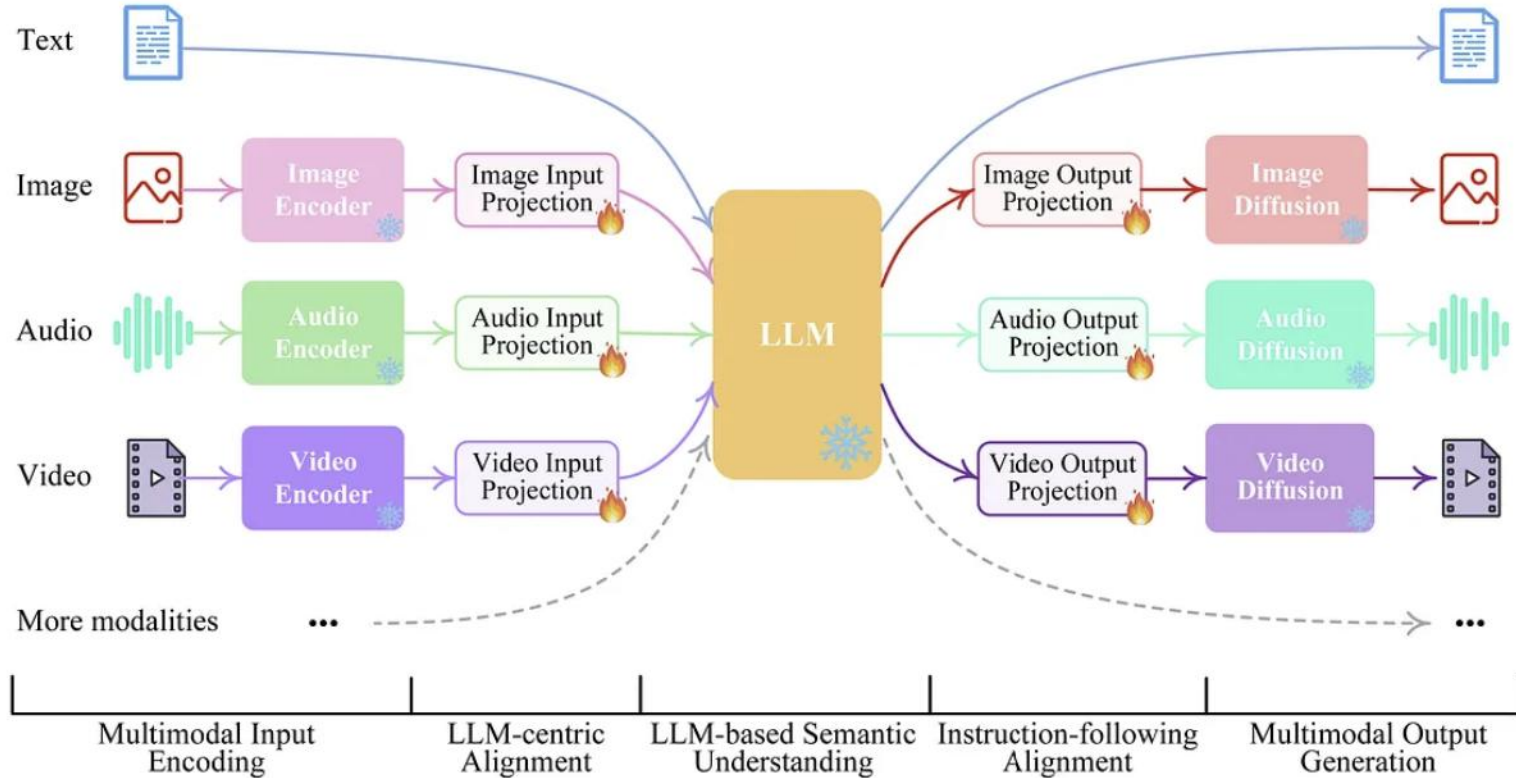


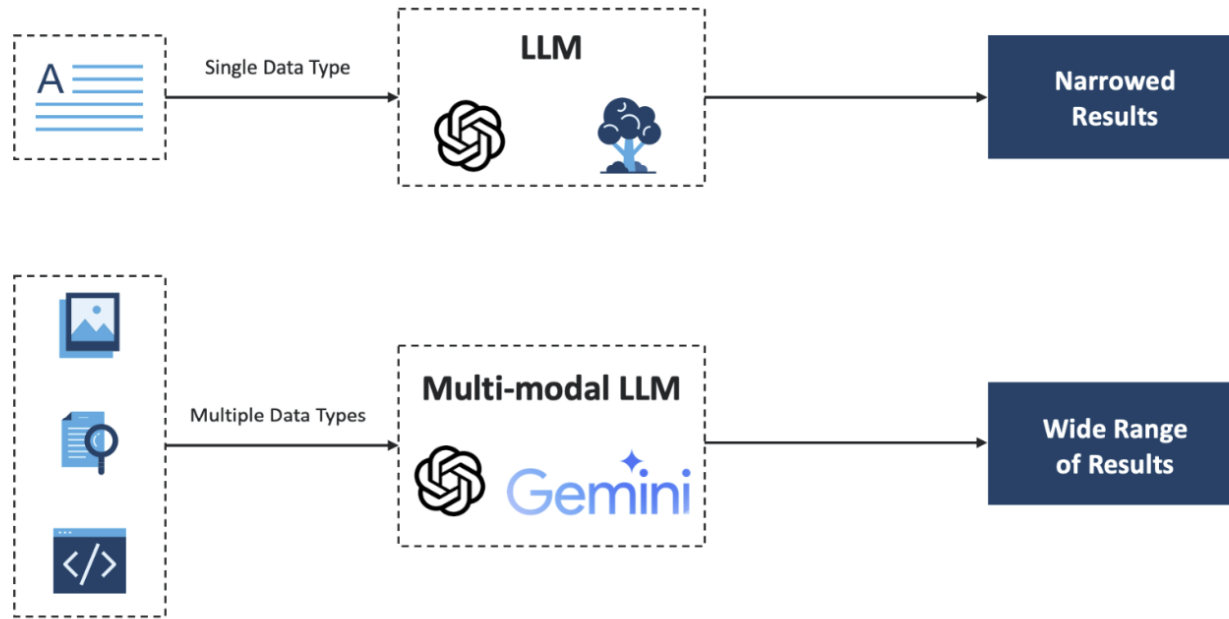
Speech to Text

Text to Speech



Multi-Modality





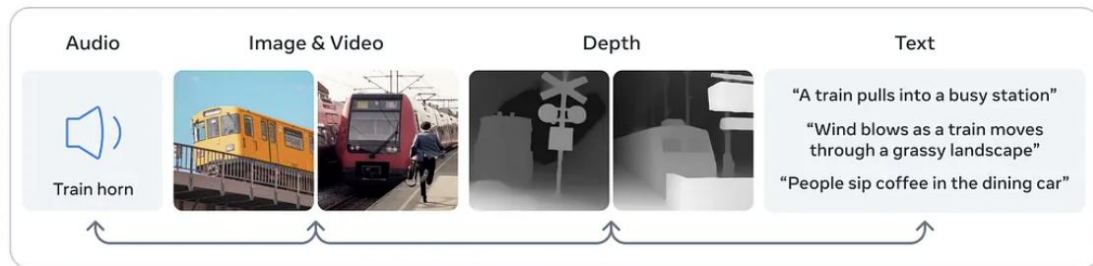
Comprensione contestuale

- Analisi simultanea di testo e immagini
- Interpretazione più accurata del contesto completo

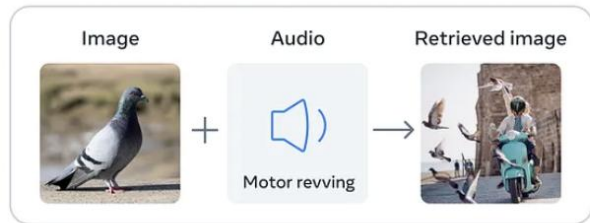
Versatilità applicativa

- Assistenza visiva in vari campi (medicina, architettura)
- Supporto decisionale basato su dati multimodali

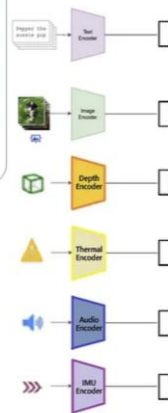
Cross-modal retrieval



Embedding-space arithmetic



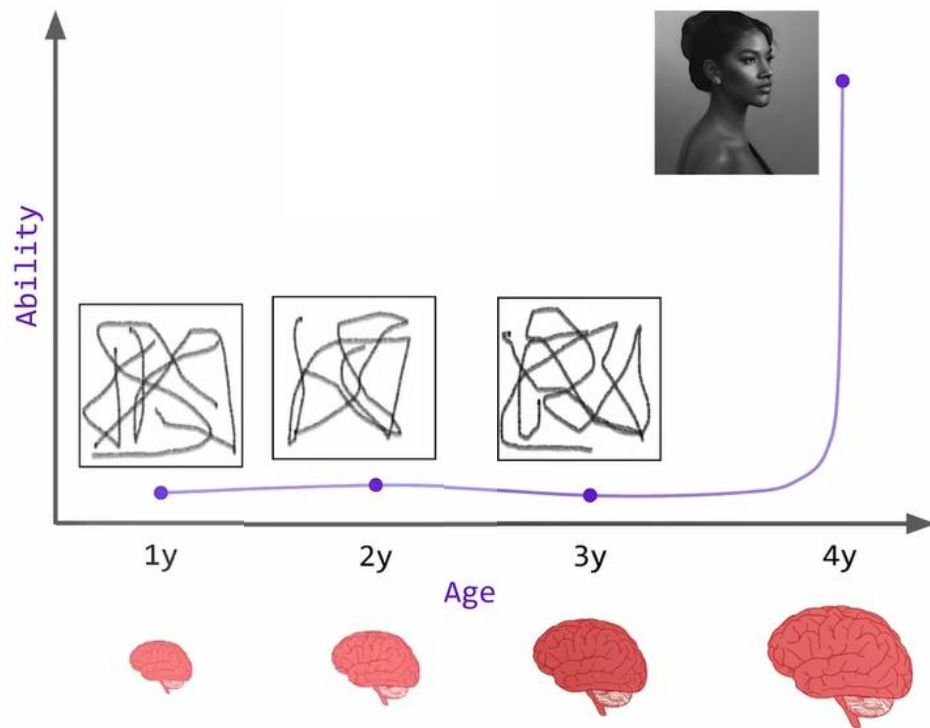
Audio to image generation



$I_1 \cdot T_1$	$I_1 \cdot T_2$	$I_1 \cdot T_3$	...	$I_1 \cdot T_N$
$I_2 \cdot T_1$	$I_2 \cdot T_2$	$I_2 \cdot T_3$	...	$I_2 \cdot T_N$
$I_3 \cdot T_1$	$I_3 \cdot T_2$	$I_3 \cdot T_3$	...	$I_3 \cdot T_N$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
$I_N \cdot T_1$	$I_N \cdot T_2$	$I_N \cdot T_3$	...	$I_N \cdot T_N$

InfoNCE loss

The Emergent Abilities of LLMs



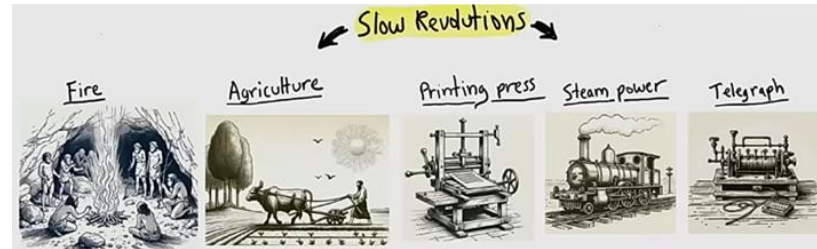
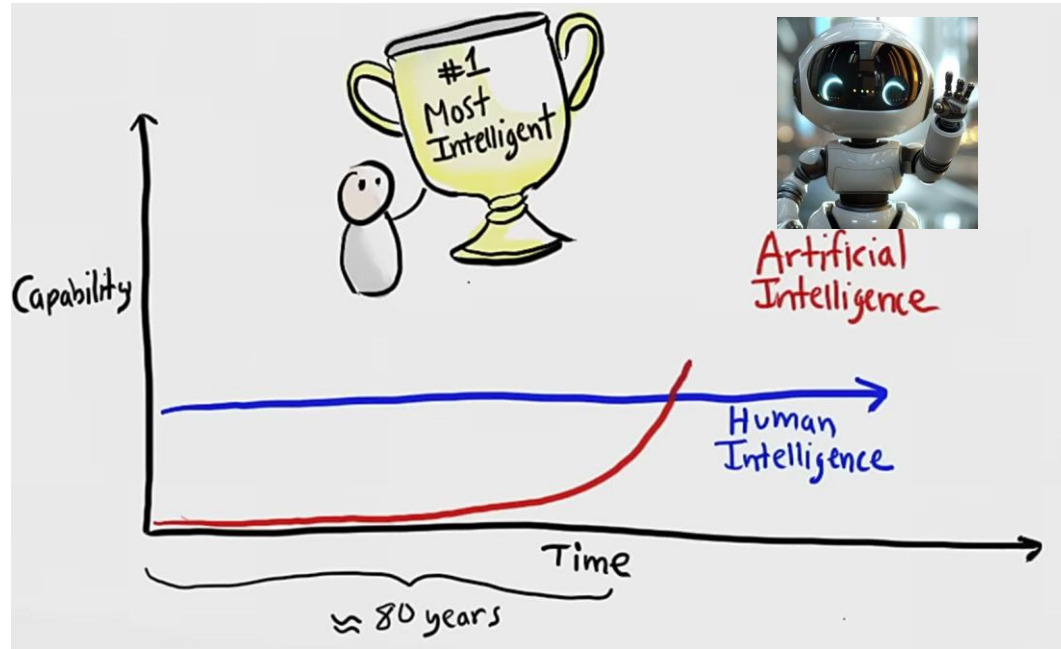
QUESTION ANSWERING

ARITHMETIC

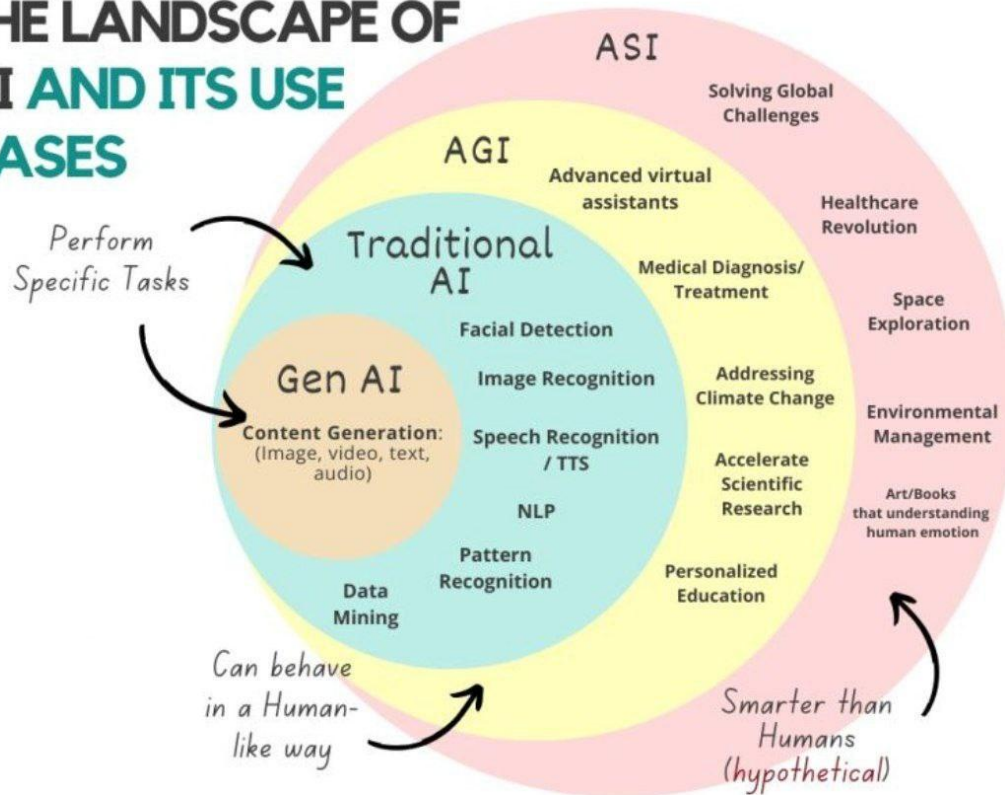


LANGUAGE UNDERSTANDING

8 billion parameters



THE LANDSCAPE OF AI AND ITS USE CASES



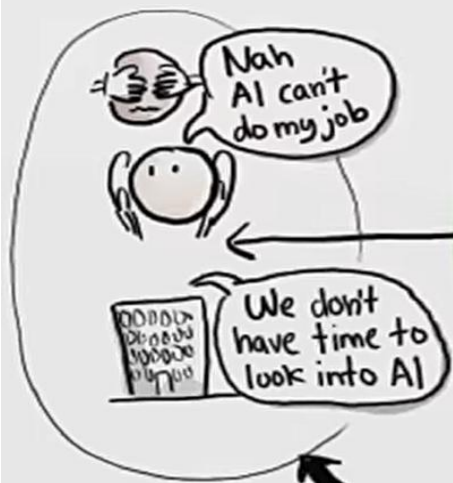
OpenAI Imagines our AI Future

Stages of Artificial Intelligence

Level 1	Chatbots, AI with conversational language
Level 2	Reasoners, human-level problem solving
Level 3	Agents, systems that can take actions
Level 4	Innovators, AI that can aid in invention
Level 5	Organizations, AI that can do the work of an organization

Mindset

Denial



Positive



Panic



AI might not take your job,
but people/companies using AI will

The role of Humans

Is human role X still needed?

doctor, developer, lawyer, CEO, teacher, etc...

Decide what to ask
and how

Evaluate results

beware of
hallucinations!

Compensate for AI weaknesses

Provide context

Legal compliance
Data security
etc, etc...

I work as X.
My biggest challenge is Y.
How can you help me?



Human



AI

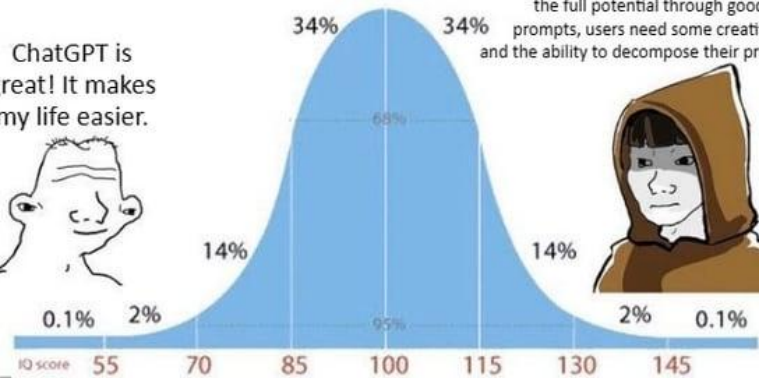
your colleague

ChatGPT is overhyped! It's just a LargeLanguageModel and therefore only capable of producing texts. In my highly complex field of work, it is no help at all. When people ask real questions to it, they will experience that it only makes things up. IT EVEN MAKES MISTAKES WITH SIMPLE MATHS!

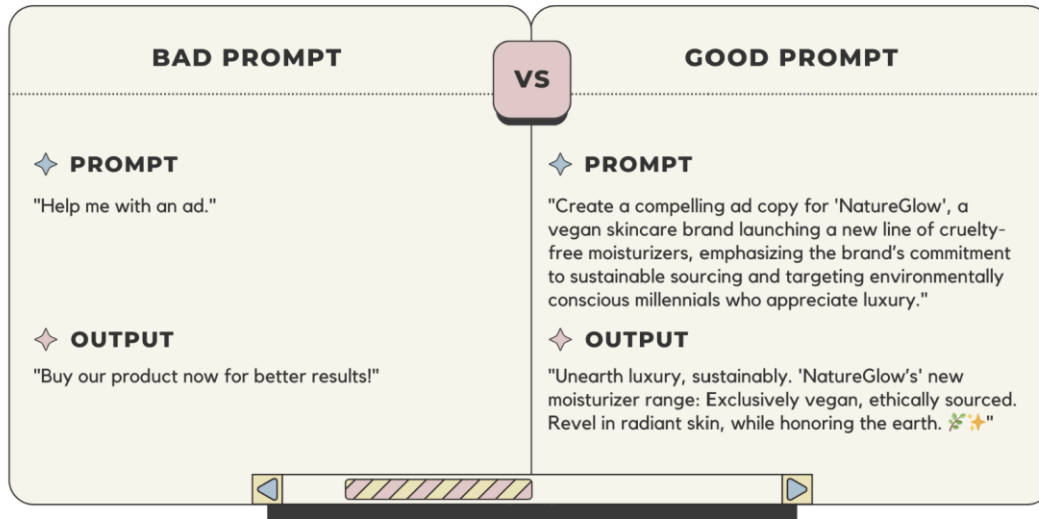


ChatGPT is a revolutionary tool with many use cases. It can sometimes make things up, therefore the results should be taken with caution like with every other non-scientific source. To extract the full potential through good prompts, users need some creativity and the ability to decompose their problems.

ChatGPT is great! It makes my life easier.



imgflip.com

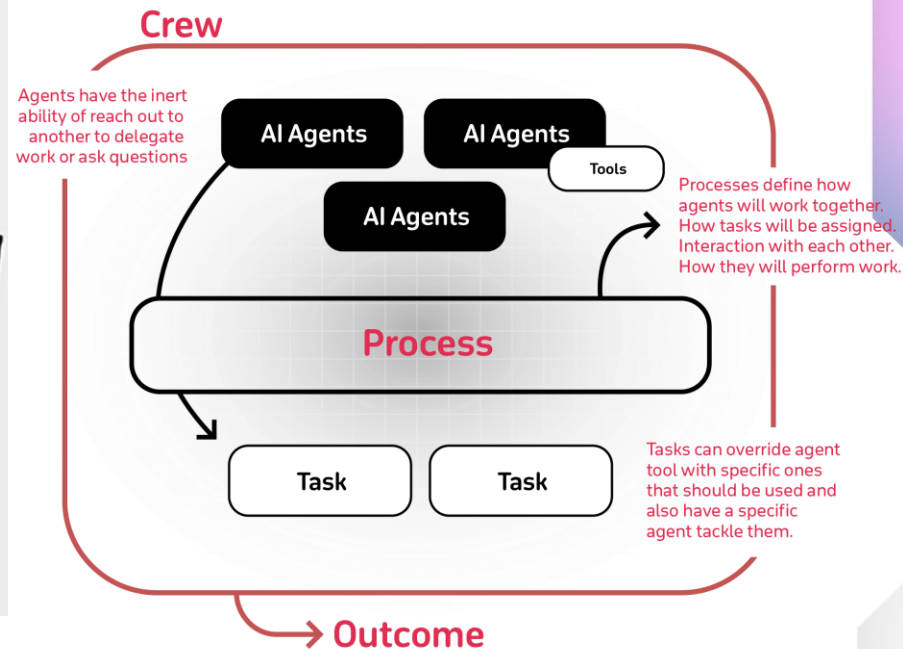


GOOD PROMPT	BAD PROMPT
• Clear • Specific • Has enough context • Contains audience insights	• Vague • Generic • Has little to no context • Doesn't have any details

Better prompt engineering skills = Better results

Side effect: Better communication skills overall

Study, Practice, Learn



Chain of Thought



Text – General description of the action



Reasoning – Why was this action taken?



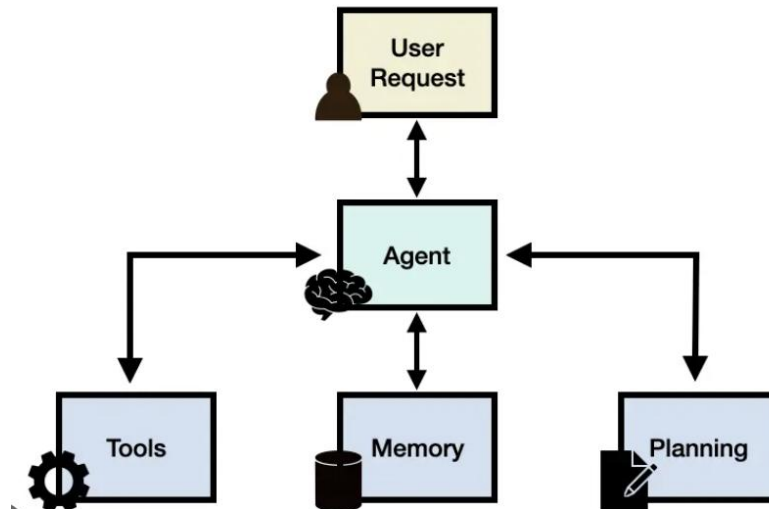
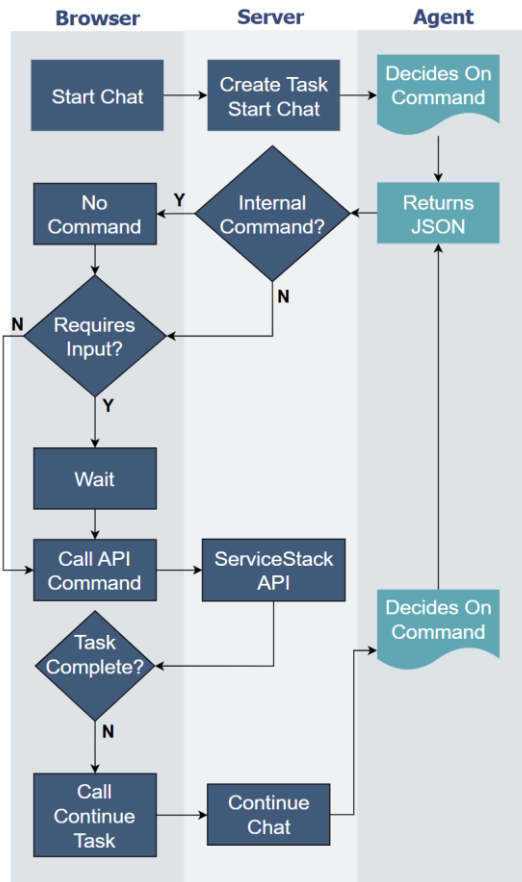
Plan – List steps along the way



Criticism – What could be a problems?



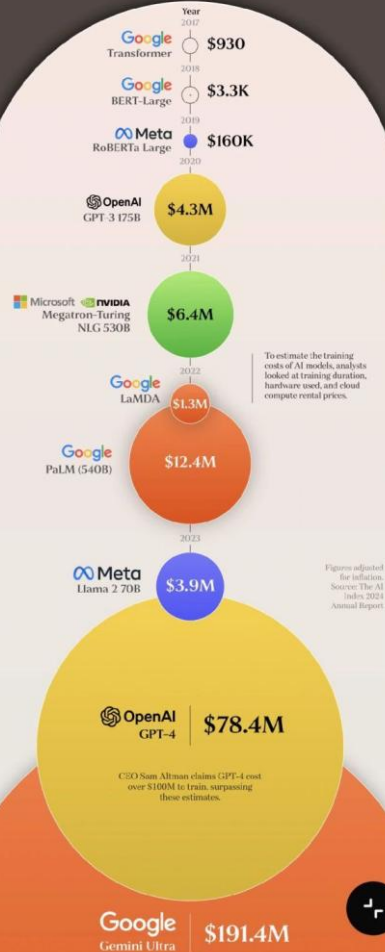
Speak – Output for the user



Generative AI Use Case Prevalence



The Cost of Training AI



Although OpenAI asked them not to, the cost of O3 was published in this chart.



Parameters (Bn) open access

anthropic chinese google meta microsoft mistral openAI other

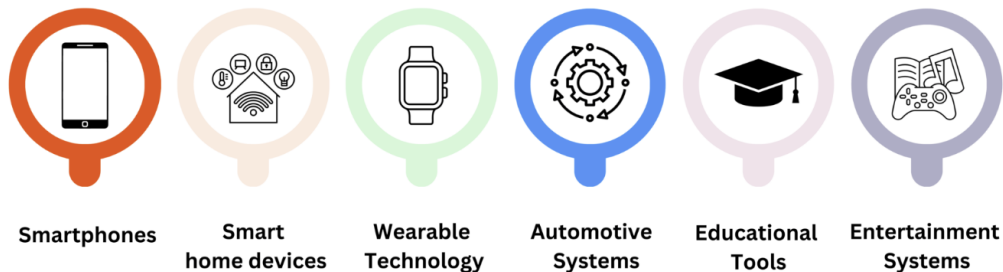


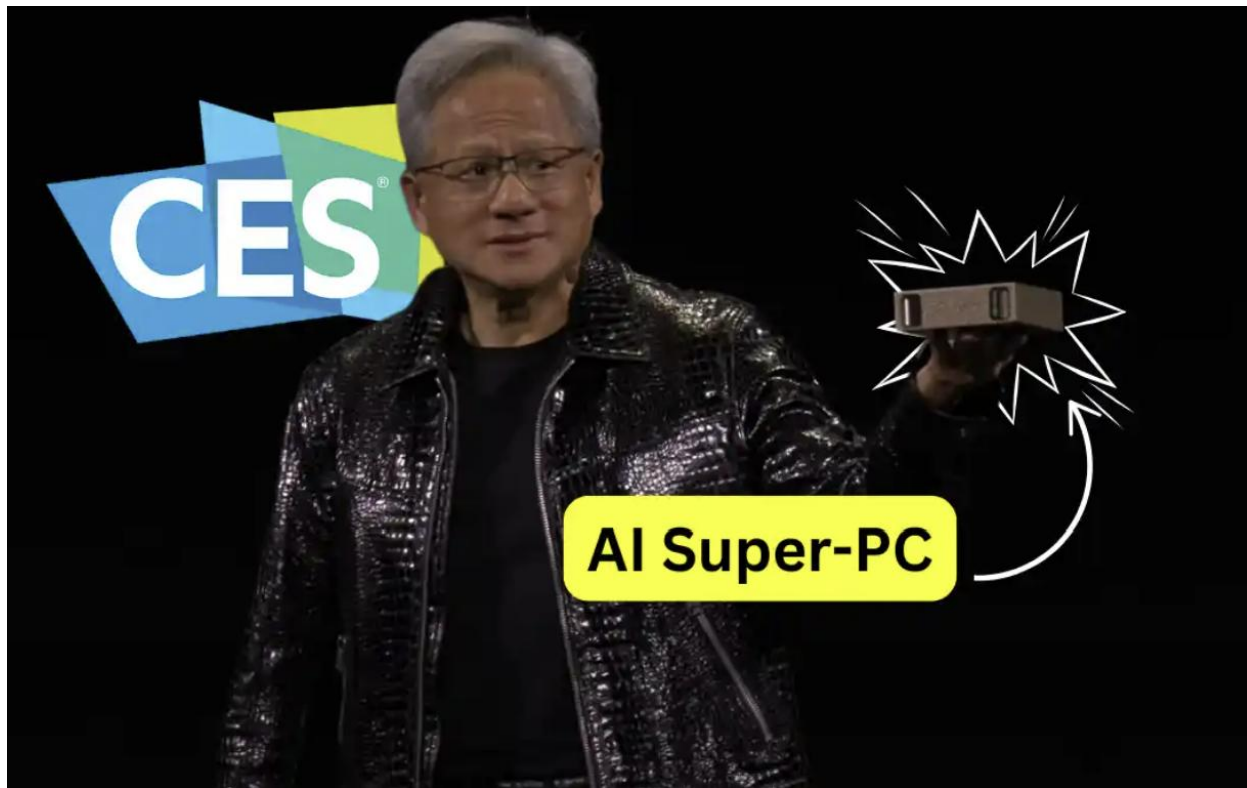
MMU = benchmark for measuring LLM capabilities
* = parameters undisclosed // source: LifeArchitect

What Makes Small Language Models so Attractive?



Potential Applications of SLMs in Technology and Services





Run 200-billion parameter LLMs locally

Phi-4 offers high quality results at a small size



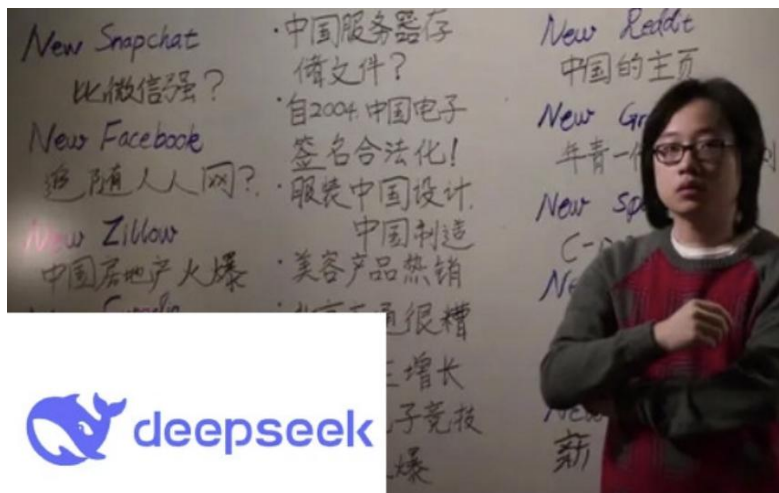


EXHIBIT 5: DeepSeek has priced their models 20-40x cheaper than OpenAI

Model	\$/M Input Tokens	\$/M Output Tokens
Deepseek-chat (V3)	\$0.14	\$0.25
gpt-4o	\$2.50	\$10.00
Δ	-94%	-98%
Deepseek-Reasoner (R1)	\$0.55	\$2.19
o1	\$15.00	\$60.00
Δ	-96%	-96%

Source: DeepSeek, OpenAI, Bernstein analysis

Benchmark (Metric)	DeepSeek-V3	Qwen2.5 72B-Inst.	Llama3.1 405B-Inst.	Claude-3.5-Sonnet-1022	GPT-4o 0513
Architecture	MoE	Dense	Dense	-	-
# Activated Params	37B	72B	405B	-	-
# Total Params	671B	72B	405B	-	-
MMLU (EM)	88.5	85.3	88.6	88.3	87.2
MMLU-Redux (EM)	89.1	85.6	86.2	88.9	88
MMLU-Pro (EM)	75.9	71.6	73.3	78	72.6
DROP (3-shot F1)	91.6	76.7	88.7	88.3	83.7
English IF-Eval (Prompt Strict)	86.1	84.1	86	86.5	84.3
GPQA-Diamond (Pass@1)	59.1	49	51.1	65	49.9
SimpleQA (Correct)	24.9	9.1	17.1	28.4	38.2
FRAMES (Acc.)	73.3	69.8	70	72.5	80.5
LongBench v2 (Acc.)	48.7	39.4	36.1	41	48.1
HumanEval-Mul (Pass@1)	82.6	77.3	77.2	81.7	80.5
LiveCodeBench(Pass@1-COT)	40.5	31.1	28.4	36.3	33.4
LiveCodeBench (Pass@1)	37.6	28.7	30.1	32.8	34.2
Code Codeforces (Percentile)	51.6	24.8	25.3	20.3	23.6
SWE Verified (Resolved)	42	23.8	24.5	50.8	38.8
Aider-Edit (Acc.)	79.7	65.4	63.9	84.2	72.9
Aider-Polyglot (Acc.)	49.6	7.6	5.8	45.3	16
AIME 2024 (Pass@1)	39.2	23.3	23.3	16	9.3
Math MATH-500 (EM)	90.2	80	73.8	78.3	74.6
CNMO 2024 (Pass@1)	43.2	15.9	6.8	13.1	10.8
Chinese CLUEWSC (EM)	90.9	91.4	84.7	85.4	87.9
C-Eval (EM)	86.5	86.1	61.5	76.7	76
C-SimpleQA (Correct)	64.1	48.4	50.4	51.3	59.3



THE BIGGEST SINGLE-DAY STOCK DECLINES

Nvidia has experienced 8 of the 10 biggest single-day stock declines.



-\$197B	Aug. 29, 2024	NVIDIA.
-\$205B	Jul. 24, 2024	NVIDIA.
-\$205B	Jul. 17, 2024	NVIDIA.
-\$206B	Apr. 29, 2022	amazon
-\$208B	Jun. 24, 2024	NVIDIA.
-\$212B	Apr. 19, 2024	NVIDIA.
-\$228B	Jan. 7, 2025	NVIDIA.
-\$251B	Feb. 3, 2022	Meta
-\$279B	Aug. 3, 2024	NVIDIA.
-\$560B	Jan. 27, 2025	NVIDIA.

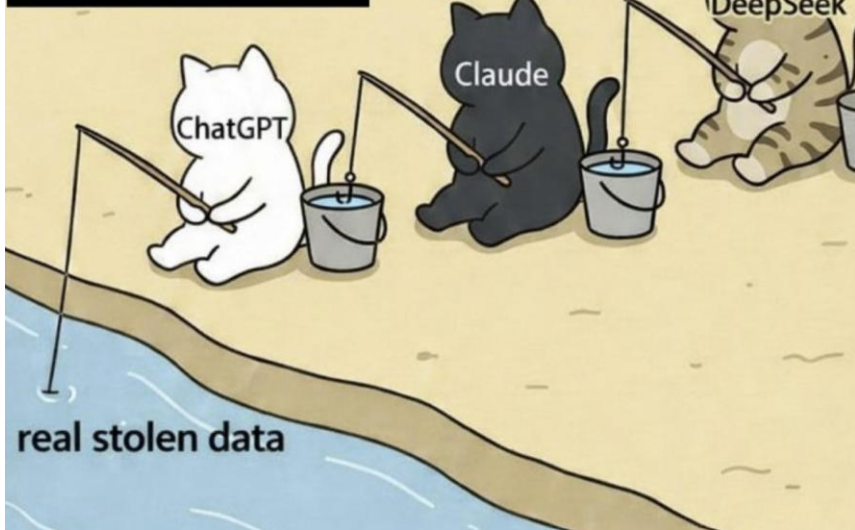
AI Anthropic  [@AnthropicAI](#) [X.com](#)

We've identified industrial-scale distillation attacks on our models by DeepSeek, Moonshot AI, and MiniMax.

These labs created over 24,000 fraudulent accounts and generated over 16 million exchanges with Claude, extracting its capabilities to train and improve their own models.

10:15 AM · 2/23/26 · 7.8M Views

3.8K 7.5K 32K 6.9K





Hugging Face

Tasks Libraries Datasets Languages Licenses Other

Q Filter Tasks by name

Multimodal

- Text-to-Image
- Image-to-Text
- Text-to-Video
- Visual Question Answering
- Document Question Answering
- Graph Machine Learning

Computer Vision

- Depth Estimation
- Image Classification
- Object Detection
- Image Segmentation
- Image-to-Image
- Unconditional Image Generation
- Video Classification
- Zero-Shot Image Classification

Natural Language Processing

- Text Classification
- Token Classification
- Table Question Answering
- Question Answering

Models 469,541 Filter by name

meta-llama/Llama-2-70b

- Text Generation • Updated 4 days ago • 25.2k

stabilityai/stable-diffusion-xl-base

- Updated 6 days ago • 2.01k • 393

openchat/openchat

- Text Generation • Updated 2 days ago • 1.3k •

llyasviel/ControlNet-v1-1

- Updated Apr 26 • 1.87k

cerspense/zeroscope_v2_XL

- Updated 3 days ago • 2.66k • 334

meta-llama/Llama-2-13b

- Text Generation • Updated 4 days ago • 328 • 64

USE CASE1:

789012 Bonifico IT60X0542811101000000123456 Mario Rossi Via Roma 123 20100 Milano importo EUR 1500,00 da ta 05/01/2024 IT60X0542811101000000123456 789012 Via Roma 123 20100
EUR 2300,50 IT72Y0300203280000400145678 Giuseppe Verdi Via Dante 45 50123 Firenze causale: pagamento fat tura data 06/01/2024 456789 IT72Y0300203280000400145678 456789 Via Dante 45 50123
bonifico eseguito 750,25EUR IT98Z0336902000001234567890 Luigi Bianchi 123456 Corso Italia 78 30172 Venez ia data 07/01/2024 IT98Z0336902000001234567890 123456 Corso Italia 78 30172
Addebito diretto SDD Core IT41X0200805364000105917552 Via Garibaldi 256 00187 987654 Roma Anna Neri comm issioni 2,00EUR IT41X0200805364000105917552 987654 Via Garibaldi 256 00187



Data	Importo	IBAN	Cod. Cliente	Ordinante	Indirizzo	CAP	Città
05/01/2024	€1.500,00	IT60X0542811101000000123456	789012	Mario Rossi	Via Roma 123	20100	Milano
06/01/2024	€2.300,50	IT72Y0300203280000400145678	456789	Giuseppe Verdi	Via Dante 45	50123	Firenze
07/01/2024	€750,25	IT98Z0336902000001234567890	123456	Luigi Bianchi	Corso Italia 78	30172	Venezia
08/01/2024	€2,00	IT41X0200805364000105917552	987654	Anna Neri	Via Garibaldi 256	00187	Roma

USE CASE2:

Classificazione completata:

	name	year	Price	kms_driven	fuel_type
0	Hyundai Santro Xing XO eRLX Euro III	2007	800.0	45000	Petrol
1	Mahindra Jeep CL550 MDI	2006	4250.0	40	Diesel
2	Hyundai Grand i10 Magna 1.2 Kappa VTVT	2014	3250.0	28000	Petrol
3	Ford EcoSport Titanium 1.5L TDCi	2014	5750.0	36000	Diesel
4	Ford Figo	2012	1750.0	41000	Diesel
...	...	...	...	...	...
812	Toyota Corolla Altis	2009	3000.0	132000	Petrol
813	Tata Zest XM Diesel	2018	2600.0	27000	Diesel
814	Mahindra Quanto C8	2013	3900.0	40000	Diesel
815	Honda Amaze 1.2 E i VTEC	2014	1800.0	50000	Petrol
816	Chevrolet Sail 1.2 LT ABS	2014	1600.0	50000	Petrol

$$R^2 = 0.23$$

LLM
Classifier

Classificazione completata:

name	vehicle_type
Hyundai Santro Xing XO eRLX Euro III	NotClassified
Mahindra Jeep CL550 MDI	Off-road
Hyundai Grand i10 Magna 1.2 Kappa VTVT	Hatchback
Ford EcoSport Titanium 1.5L TDCi	SUV
Ford Figo	Hatchback
...	...
Toyota Corolla Altis	Sedan
Tata Zest XM Diesel	Sedan
Mahindra Quanto C8	SUV
Honda Amaze 1.2 E i VTEC	Sedan
Chevrolet Sail 1.2 LT ABS	Sedan

$$R^2 = 0.86$$



MODELLI AUTOREGRESSIVI

La

PASSAGGI LOGICI ERRATI ⚠

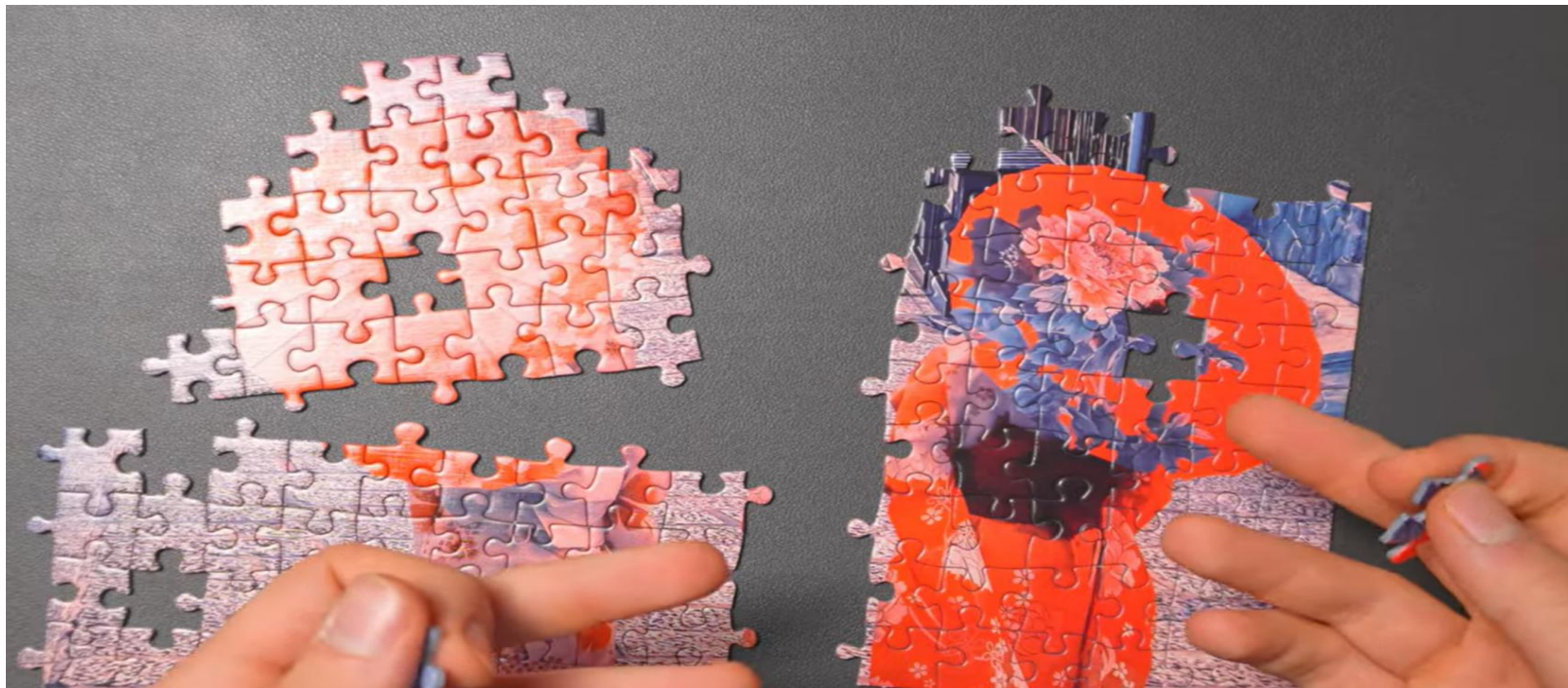
"Se ChatGPT si accorge di un errore, potrebbe tornare indietro e sistemare?"

La risposta purtroppo è NO!!!



Il s.

DIFFUSION MODELS



AUTOREGR.

DIFFUSIONE

OLTRE L'ALLUCINAZIONE: I LIMITI SISTEMICI



The diagram features a stylized iceberg floating in a dark blue sea. The tip of the iceberg, which is above the water line, is labeled 'Output Testuale (Apparente Ragionevolezza)'. The much larger part of the iceberg is submerged below the water line and is divided into three horizontal layers. The top layer is labeled 'Opacità (Black Box)' and contains the text 'Manca un sillogismo verificabile. Non esiste una catena argomentativa trasparente che un giudice possa valutare.' The middle layer is labeled 'Bias nei Dati' and contains the text 'I modelli riflettono i pregiudizi storici dei corpus di addestramento, riproducendo disparità sociali in veste tecnica.' The bottom layer is labeled 'Limiti Temporali e Geografici' and contains the text 'Knowledge Cutoff (ignora sentenze recenti) e Bias Giurisdizionale (affidabilità inferiore nel diritto italiano rispetto al Common Law).' The background is a dark blue gradient with some light blue geometric shapes on the left and right sides.

Output Testuale
(Apparente Ragionevolezza)

Opacità (Black Box)

Manca un sillogismo verificabile. Non esiste una catena argomentativa trasparente che un giudice possa valutare.

Bias nei Dati

I modelli riflettono i pregiudizi storici dei corpus di addestramento, riproducendo disparità sociali in veste tecnica.

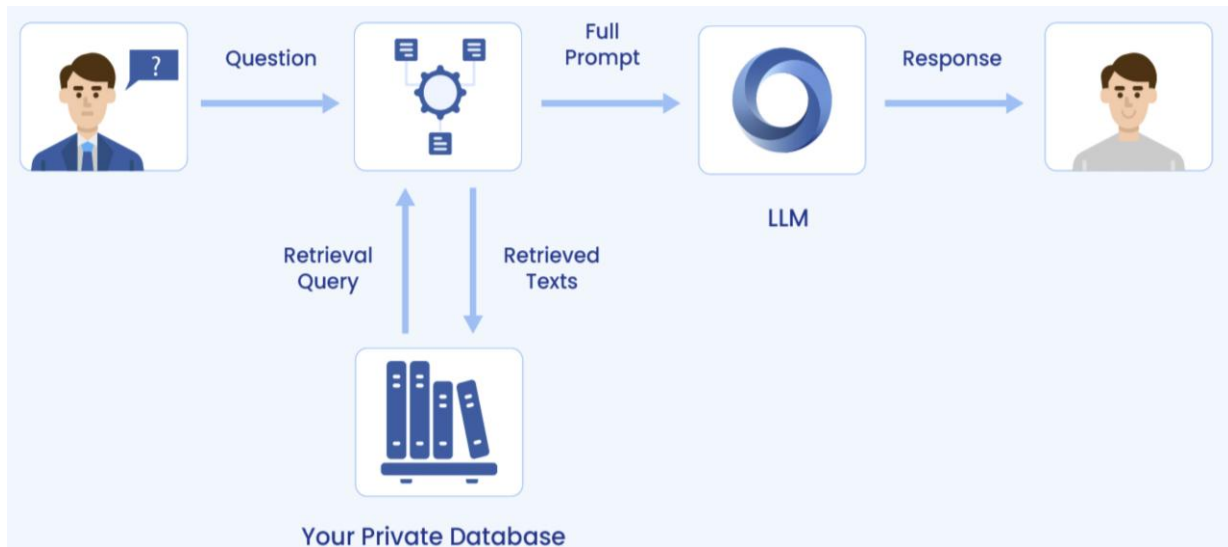
Limiti Temporali e Geografici

Knowledge Cutoff (ignora sentenze recenti) e Bias Giurisdizionale (affidabilità inferiore nel diritto italiano rispetto al Common Law).

ChatGPT (KB-Interna)



Sistema RAG (KB-Esterna)



Grazie

youremail@freepik.com

+34 654 321 432

yourwebsite.com



Synthetic data for AI and machine learning is **more flexible** than real data

“Synthetic data generation accelerates the analytics development cycle, lessens regulatory concerns and lowers the cost of data acquisition.” – Gartner

Synthetic data is the perfect fuel that AI and machine learning development projects need. Synthetic data is created by generative AI algorithms, which can be instructed to create bigger, smaller, fairer or richer versions of the original data. Due to how the synthesization process takes place, the data can be augmented to fit certain characteristics. In a way, synthetic data is like modelling clay for data scientists and data managers. For example, upsampling minority groups in a dataset can improve the performance of machine learning models. Or human bias embedded in the original data can be removed by introducing fairness constraints to the generation process.

Value of Synthetic Data

- **Model Training:** Provides vast training data with tailored scenarios for robust models.
- **Data Sharing:** Enables collaboration without privacy concerns.
- **Bias Mitigation:** Addresses biases in real data for fairer applications.
- **Testing and Validation:** Supports comprehensive testing in controlled environments.
- **Cost Reduction:** Reduces data collection and labeling costs.



Real life **examples** of synthetic data projects

AI-generated synthetic data is used by companies accross the world to accelerate their data innovation projects in a privacy-compliant and agile way.

- ▶ Telefónica uses synthetic customer data for analytics
- ▶ Erste Bank used synthetic test data to develop a successful mobile banking app
- ▶ JPMorgan leverages a synthetic data sandbox to speed up data-intensive POCs with third party vendors
- ▶ Anthem uses synthetic data to detect fraud and for delivering personalized service to members

Method	Protection from re-identification risk	Feature Statistics	Feature correlations	ML performance
Synthetic data	High	High	High	High
Randomization	Low	Medium	Low	Low
Permutation	Low	High	Very low	Very low
Generalization	Medium	Low	Low	Low
Pseudonymization	Very low	High	High	High
Data masking	Very low	Very low	Very low	Very low

How does synthetic data compare to other data anonymization techniques?

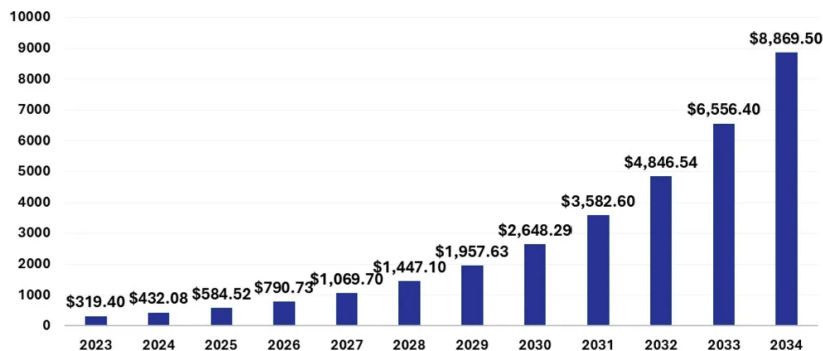
Research has shown many times that legacy data anonymization techniques endanger privacy. They often also degrade the utility of the data.

Even worse – a lot of companies are equating pseudonymization with anonymization. But from a legal perspective, pseudonymized data is still personal data. And it needs to be treated and protected as just that.

How can synthetic data help data-driven businesses?

Extract the full value of your data assets. We enable responsible privacy enhancing data availability, free of bias, and incompleteness.

Synthetic Data Generation Market Size 2023 to 2034 (USD Million)



DIPENDENZA EMOTIVA E GHOSTING: I RISCHI DELL'UTILIZZO DI CHATGPT

OpenAI mette in guardia rispetto a
possibile "antropomorfizzazione e
dipendenza emotiva" dall'IA

OpenAI ha messo in guardia gli utenti
rispetto ad alcuni **potenziali effetti
negativi dell'uso di ChatGPT**,
presentati nel rapporto '**GPT-4o
System Card**', dedicato a tutte le
analisi di sicurezza svolte prima del
rilascio di **GPT-4o**.

In particolare, i **rischi** che preoccupano maggiormente riguardano **gli effetti di un'interazione prolungata con ChatGPT:**



la possibilità che insegni agli utenti comportamenti negativi nelle relazioni interpersonali, come il ghosting



l'eventualità di portare l'utente ad una sorta di dipendenza emotiva

Perché potrebbe verificarsi tutto questo? L'**intelligenza artificiale** è una **tecnologia estremamente innovativa**, in **rapida e costante evoluzione**. Le sue applicazioni portano **enormi benefici**, **ma anche rischi da non sottovalutare**. Riuscire a **garantire** che venga **utilizzata in modo etico e responsabile** è fondamentale.

È per questo che **OpenAI** sta **collaborando con sociologici e psicologi** mentre **intensifica le sue ricerche e sviluppa nuove strategie di sicurezza**.

the commentary in snack-sized mouthfuls



"I want AI to do my laundry and dishes so that I can do art and writing, not for AI to do my art and writing so that I can do my laundry and dishes."

Author and videogame enthusiast **Joanna Maciejewska** nails it (although bathroom cleaning goes ahead of laundry and dishes)

"I'm sure I deserve a lot of criticism"